Section J - List of Attachments

The following attachments were added:

Attachment 10 - AD 22 Area Site Plan_3 12 2025
Attachment 11 - AD 23 Site Utility Plan_3 12 2025
Attachment 9 - AD 21 Conceptual Layout Plan_17 11 2025

Miscellaneous text in this section has been modified to:

Attachment 1 - Statement of Work

Attachment 2 - Wage Determination

Attachment 3 - Soil Management Plan

Attachment 4 - DD254 Contract Security Classification Checklist

Attachment 5 - Key Personnel

Attachment 6 - Hazardous Material Usage Log

Attachment 7 - Generator Usage Log

Attachment 8 - Site Visit Instructions

Attachment 9 - AD 21 Conceptual Layout Plan_17 11 2025

Attachment 10 - AD 22 Area Site Plan_3 12 2025

Attachment 11 - AD 23 Site Utility Plan_3 12 2025

NOTE 16: RETIREMENT PLANS (Continued)

Arrow's pension plan weighted-average asset allocations at December 31, 2005, and 2004, by asset category are as follows:

	Plan Assets At December 31,	
	2005	2004
Asset Category:		
Cash	**1.0%**	0.6%
Mortgages	**0.3**	0.5
Debt Securities	**---**	1.0
Company Stock	**5.3**	5.8
Mutual Funds – Equity	**73.8**	72.4
Mutual Funds – Fixed Income	**19.6**	19.7
Total	**100.0%**	100.0%

At December 31, 2005 and 2004, plan assets included shares of mutual funds advised by Arrow's subsidiary, North Country Investment Advisers, Inc., with a market value of $16,846 and $17,676, respectively. At December 31, 2005 and 2004, plan assets also included 51 and 47 shares, respectively, of Arrow Financial Corporation common stock with a market value of $1,334 and $1,457, respectively. During the respective years, the plan received $46 and $40 from cash dividends on Arrow's common stock. In accordance with ERISA guidelines, the Board authorized the purchase of Arrow Financial Corp. common stock up to 10% of the fair market value of the plan's assets at the time of acquisition.

Pension Plan Investment Policies and Strategies:

Return Requirements:

The portfolio should achieve an inflation-protected rate of return at least equal to the actuarial assumption of 8.75%.

Risk Tolerance:

The Plan has the flexibility to accept an average to above-average degree of risk. Key factors to consider in reaching conclusions regarding risk tolerance are: (i) the pension plan must meet ERISA prudence requirements, which apply to the entire portfolio, not just its individual component securities, (ii) the Plan's two key actuarial assumptions with regard to expected long-term return on plan assets and salary progression are 8.75% and 3.50%, respectively. The expected long-term return on plan assets is reasonable relative to historic results over the last ten years of 8.95%. The salary progression rate is in line with past results, (iii) the plan is valued annually, (iv) Arrow's average employee age is reasonably low (43.0), and the time horizon is long, and (v) the Plan's operating results have been relatively strong and consistent.

Asset Allocation:

The Plan's limited liquidity requirements permit a low level of short-term reserves, which in any event do not meet the plan's 8.75% return requirement. All of the constraints suggest that moderate emphasis on common stocks is appropriate. With historically low interest rates, a lower weighting in bonds is appropriate. A separate asset allocation policy is reviewed by the Board on a regular basis and documented.

Investment Strategy:

The equity portion of the plan will be invested in a diversified portfolio of equity securities of companies with small, mid, and large capitalizations. Both domestic and international equities are allowed. While the plan is allowed to invest in the common stock of Arrow Financial up to 10% of the fair market value of plan assets at purchase, the plan assets will not be concentrated in any particular industry. Both growth and value styles will be employed to increase the diversification and offer varying opportunities for appreciation.

NOTE 16: RETIREMENT PLANS (Continued)

The fixed income portion of the plan will be invested in U.S. dollar denominated investment grade bonds or debt securities. Portfolio securities shall be rated within the top four ratings categories by nationally recognized ratings agencies such as Moody's and Standard & Poor's. The bond portfolio will maintain a dollar-weighted average quality of "A" or better and an average dollar-weighted maturity between 1 and 10 years. The bond portion will be invested without regard to industry or sector based on analysis of each target security's structural and repayment features, current pricing and trading opportunities as well as credit quality of the issuer. Bonds with ratings that fall below the portfolio's rating requirements will be sold only when it is in the best interests of the plan.

Cash Flows

Based on the funded status at December 31, 2005, Arrow does not expect to make a contribution to the qualified defined benefit pension plan during 2006. Arrow makes contributions for its postretirement benefits in an amount equal to actual expenses for the year. That amount is estimated to be $350 for 2006.

NOTE 17: OTHER EMPLOYEE BENEFIT PLANS (In Thousands)

Arrow maintains an employee stock ownership plan ("ESOP"). Substantially all employees of Arrow and its subsidiaries are eligible to participate upon satisfaction of applicable service requirements. The ESOP borrowed $105, $464 and $853 in 2001, 2000 and 1999, respectively, from one of Arrow's subsidiary banks to purchase outstanding shares of Arrow's common stock. The notes require annual payments of principal and interest through 2011. Arrow's ESOP expense amounted to $196, $500 and $450 in 2005, 2004 and 2003, respectively. As the debt is repaid, shares are released from collateral based on the proportion of debt paid to total debt outstanding for the year and allocated to active employees.

Shares pledged as collateral are reported as unallocated ESOP shares in shareholders' equity. As shares are released from collateral, Arrow reports compensation expense equal to the current average market price of the shares, and the shares become outstanding for earnings per share computations. The ESOP shares as of December 31, 2005 were as follows:

Allocated Shares	737
Shares Released for Allocation During 2005	14
Unallocated Shares	82
Total ESOP Shares	833
Market Value of Unallocated Shares	$2,153

Under the employee stock purchase plan ("ESPP"), employees may purchase shares of Arrow's common stock, up to $24 annually, at a discount to the prevailing market price (currently a 5% discount). Under the ESPP, shares are issued by Arrow without a charge to earnings in accordance with SFAS No. 123. Substantially all employees of Arrow and its subsidiaries are eligible to participate upon satisfaction of applicable service requirements.

Arrow also sponsors a Short-Term Incentive Award Plan for senior management and a Profit Sharing Plan for substantially all employees. The combined cost of these plans was $323, $667 and $525 for 2005, 2004 and 2003, respectively.

NOTE 18: STOCK OPTION PLANS

Arrow has established fixed Incentive Stock Option and Non-qualified Stock Option Plans. At December 31, 2005, approximately 221,000 shares remained available for grant under these plans. Options may be granted at a price no less than the greater of the par value or fair market value of such shares on the date on which such option is granted, and generally expire ten years from the date of grant. The options usually vest over a four-year period, however, in December 2005 Arrow's Compensation Committee of the Board of Directors accelerated the vesting for all the remaining unvested shares from stock options granted in 2002 through 2004. The action to accelerate the vesting of the stock options was made to eliminate the non-cash compensation expense that would otherwise have been recognized by the Company in the 2006-2008 period due to the required adoption of FASB Statement 123(R) on January 1, 2006.

A summary of the status of Arrow's stock option plans as of December 31, 2005, 2004 and 2003 and changes during the years then ended is presented below (all share and per share data have been adjusted for stock splits and dividends, including the September 2005 3% stock dividend).

NOTE 18: STOCK OPTION PLANS (Continued)

Options:	**2005** Shares	**2005** Weighted-Average Exercise Price	2004 Shares	2004 Weighted-Average Exercise Price	2003 Shares	2003 Weighted-Average Exercise Price
Outstanding at January 1	**636,482**	**$18.34**	669,048	$15.83	657,925	$14.44
Granted	**---**	**---**	65,920	31.09	66,412	26.24
Exercised	**(98,723)**	**9.93**	(98,074)	9.82	(55,147)	11.66
Forfeited	**(615)**	**28.68**	(412)	25.92	(142)	20.32
Outstanding at December 31	**537,144**	**19.87**	636,482	18.33	669,048	15.84
Exercisable at December 31	**537,144**	**19.87**	463,494	15.00	484,518	12.97

The following table summarizes information about Arrow's stock options at December 31, 2005:

	Options Outstanding			Options Exercisable	
Range of Exercise Prices	Number Outstanding At 12/31/05	Weighted-Average Remaining Contractual Life	Weighted-Average Exercise Price	Number Exercisable at 12/31/05	Weighted-Average Exercise Price
$11.00-$11.99	46,449	0.9	$11.34	46,449	$11.34
$12.00-$15.99	157,222	3.8	13.70	157,222	13.70
$16.00-$19.99	59,993	1.9	16.32	59,993	16.32
$20.00-$23.99	73,544	6.0	20.39	73,544	20.39
$24.00-$27.99	134,428	7.4	25.89	134,428	25.89
$28.00-$32.00	65,508	8.6	31.09	65,508	31.09
	537,144	5.2	19.87	537,144	19.87

NOTE 19: SHAREHOLDER RIGHTS PLAN

In 1997, the Board of Directors of Arrow adopted a shareholder rights plan. The plan provides for the distribution of one preferred stock purchase right for each outstanding share of common stock of Arrow. Each right entitles the holder, following the occurrence of certain events, to purchase a unit consisting of one-hundredth of a share of Series 1 Junior Participating Preferred Stock, at a purchase price of $35.51 (adjusted for stock dividends and stock splits) per unit, subject to adjustment. The rights will not be exercisable or transferable apart from the common stock except under certain circumstances in which a person or group of affiliated persons acquires, or commences a tender offer to acquire, 20% or more of Arrow's common stock. Rights held by such an acquiring person or persons may thereafter become void. Under certain circumstances a right may become a right to purchase common stock or assets of Arrow or common stock of an acquiring corporation at a substantial discount. Under certain circumstances, Arrow may redeem the rights at $.01 per right. The rights will expire in April 2007 unless earlier redeemed or exchanged by Arrow.

NOTE 20: OTHER OPERATING EXPENSE (In Thousands)

Other operating expenses included in the consolidated statements of income are as follows:

	2005	2004	2003
Legal and Other Professional Fees	**$1,393**	$1,225	$1,229
Computer Services	**1,270**	1,365	1,234
Postage	**1,196**	1,090	1,097
Stationery and Printing	**1,035**	864	933
Telephone and Communications	**735**	716	750
Advertising and Promotion	**700**	639	672
Intangible Asset Amortization (see Notes 1 and 8)	**386**	42	37
FDIC and Other Insurance	**359**	356	344
Charitable Contributions	**164**	170	160
All Other	**1,469**	1,338	1,764
Total Other Operating Expense	**$8,707**	$7,805	$8,220

NOTE 21: INCOME TAXES (In Thousands)

The provision for income taxes is summarized below:

	2005	2004	2003
Current Tax Expense:			
Federal	**$7,134**	$7,433	$5,940
State	**779**	929	435
Total Current Tax Expense	**7,913**	8,362	6,375
Deferred Tax Expense			
Federal	**154**	531	1,817
State	**36**	66	414
Total Deferred Tax Expense	**190**	597	2,231
Total Provision for Income Taxes	**$8,103**	$8,959	$8,606

The provisions for income taxes differed from the amounts computed by applying the U.S. Federal Income Tax Rate of 35% for 2005, 2004 and 2003 to pre-tax income as a result of the following:

	2005	2004	2003
Computed Tax Expense at Statutory Rate	**$9,360**	$9,953	$9,633
Increase (Decrease) in Income Taxes Resulting From:			
Tax-Exempt Income	**(1,757)**	(1,747)	(1,685)
Nondeductible Interest Expense	**151**	119	130
State Taxes, Net of Federal Income Tax Benefit	**533**	647	552
Other Items, Net	**(184)**	(13)	(24)
Total Provision for Income Taxes	**$8,103**	$8,959	$8,606

The tax effects of temporary differences that give rise to significant portions of the deferred tax assets and deferred tax liabilities at December 31, 2005 and 2004 are presented below:

	2005	2004
Deferred Tax Assets:		
Allowance for Loan Losses	**$ 4,761**	$4,687
Pension and Deferred Compensation Plans	**3,160**	3,061
Minimum Pension Liability	**528**	232
Net Unrealized Losses on Securities Available-for-Sale	**2,498**	---
Other	**272**	235
Total Gross Deferred Tax Assets	**11,219**	8,215
Deferred Tax Liabilities:		
Pension Plans	**3,123**	3,003
Depreciation	**709**	818
Deferred Income	**2,473**	2,398
Net Unrealized Gains on Securities Available-for-Sale	**---**	517
Goodwill	**1,677**	1,352
Other	**15**	26
Total Gross Deferred Tax Liabilities	**7,997**	8,114
Net Deferred Tax Assets	**$3,222**	$ 101

Management believes that the realization of the recognized net deferred tax assets at December 31, 2005 and 2004 is more likely than not, based on existing loss carryback ability, available tax planning strategies and expectations as to future taxable income. Accordingly, there was no valuation allowance for deferred tax assets as of December 31, 2005 and 2004.

NOTE 22: LEASE COMMITMENTS (In Thousands)

At December 31, 2005, Arrow was obligated under a number of noncancelable operating leases for buildings and equipment. Certain of these leases provide for escalation clauses and contain renewal options calling for increased rentals if the lease is renewed.

Future minimum lease payments on operating leases at December 31, 2005 were as follows:

	Operating Leases
2006	$ 267
2007	269
2008	266
2009	218
2010	219
Later Years	1,524
Total Minimum Lease Payments	$2,763

Arrow leases two of its branch offices, at market rates, from Stewart's Shops Corp. Gary Dake, president of Stewart's Shops Corp., serves on both the boards of Arrow and Saratoga National Bank and Trust Company.

NOTE 23: FINANCIAL INSTRUMENTS WITH OFF-BALANCE SHEET RISK AND CONTINGENT LIABILITIES (In Thousands)

Arrow is party to financial instruments with off-balance sheet risk in the normal course of business to meet the financing needs of its customers. These financial instruments include commitments to extend credit and standby letters of credit. Commitments to extend credit include home equity lines of credit, commitments for residential and commercial construction loans and other personal and commercial lines of credit. Those instruments involve, to varying degrees, elements of credit and interest rate risk in excess of the amount recognized in the consolidated balance sheets. The contract or notional amounts of those instruments reflect the extent of the involvement Arrow has in particular classes of financial instruments.

Arrow's exposure to credit loss in the event of nonperformance by the other party to the financial instrument for commitments to extend credit and standby letters of credit is represented by the contractual notional amount of those instruments. Arrow uses the same credit policies in making commitments and conditional obligations as it does for on-balance sheet instruments.

Commitments to extend credit were $142,299 and $138,338 at December 31, 2005 and 2004, respectively. These commitments are agreements to lend to a customer as long as there is no violation of any condition established in the contract. Commitments generally have fixed expiration dates or other termination clauses and may require payment of a fee. Since many of the commitments are expected to expire without being drawn upon, the total commitment amounts do not necessarily represent future cash requirements. Arrow evaluates each customer's creditworthiness on a case-by-case basis. Home equity lines of credit are secured by residential real estate. Construction lines of credit are secured by underlying real estate. For other lines of credit, the amount of collateral obtained, if deemed necessary by Arrow upon extension of credit, is based on management's credit evaluation of the counterparty. Collateral held varies, but may include accounts receivable, inventory, property, plant and equipment, and income-producing commercial properties. Most of the commitments are variable rate instruments.

In November 2002, the FASB issued FASB Interpretation No. 45 ("FIN No. 45"), "Guarantor's Accounting and Disclosure Requirements for Guarantees, Including Indirect Guarantees of Indebtedness of Others; an Interpretation of FASB Statements No. 5, 57, and 107 and rescission of FASB Interpretation No. 34." FIN No. 45 requires certain new disclosures and potential liability-recognition for the fair value at issuance of guarantees that fall within its scope. Under FIN No. 45, Arrow does not issue any guarantees that would require liability-recognition or disclosure, other than its standby letters of credit.

NOTE 23: FINANCIAL INSTRUMENTS WITH OFF-BALANCE SHEET RISK AND CONTINGENT LIABILITIES (Continued)

Arrow has issued conditional commitments in the form of standby letters of credit to guarantee payment on behalf of a customer and guarantee the performance of a customer to a third party. Standby letters of credit generally arise in connection with lending relationships. The credit risk involved in issuing these instruments is essentially the same as that involved in extending loans to customers. Contingent obligations under standby letters of credit totaled $2,984 and $2,630 at December 31, 2005 and 2004, respectively, and represent the maximum potential future payments Arrow could be required to make. Typically, these instruments have terms of 12 months or less and expire unused; therefore, the total amounts do not necessarily represent future cash requirements. Each customer is evaluated individually for creditworthiness under the same underwriting standards used for commitments to extend credit and on-balance sheet instruments. Company policies governing loan collateral apply to standby letters of credit at the time of credit extension. Loan-to-value ratios will generally range from 50% for movable assets, such as inventory, to 100% for liquid assets, such as bank CD's. Fees for standby letters of credit range from 1% to 3% of the notional amount. Fees are collected upfront and amortized over the life of the commitment. The carrying amount and fair value of Arrow's standby letters of credit at December 31, 2005 and 2004 were insignificant. The fair value of standby letters of credit is based on the fees currently charged for similar agreements or the cost to terminate the arrangement with the counterparties.

Under SFAS No. 107 the fair value of commitments to extend credit is determined by estimating the fees to enter into similar agreements, taking into account the remaining terms and present creditworthiness of the counterparties, and for fixed rate loan commitments, the difference between the current and committed interest rates. Arrow provides several types of commercial lines of credit and standby letters of credit to its commercial customers. The pricing of these services is not isolated as Arrow considers the customer's complete deposit and borrowing relationship in pricing individual products and services. The commitments to extend credit also include commitments under home equity lines of credit, for which Arrow charges no fee. The carrying value and fair value of commitments to extend credit are not material and Arrow does not expect to incur any material loss as a result of these commitments.

In the normal course of business, Arrow and its subsidiary banks become involved in a variety of routine legal. At present, there are no legal proceedings pending or threatened, which in the opinion of management and counsel, would result in a material loss to Arrow.

NOTE 24: FAIR VALUE OF FINANCIAL INSTRUMENTS (In Thousands)

The following table presents a summary at December 31 of the carrying amount and fair value of Arrow’s financial instruments not carried at fair value or an amount approximating fair value:

	2005		2004	
	Carrying Amount	**Fair Value**	Carrying Amount	Fair Value
Securities Held-to-Maturity (Note 3)	**$118,123**	**$118,495**	$108,117	$111,058
Net Loans (Note 4)	**984,304**	**971,835**	863,265	869,252
Time Deposits (Note 9)	**375,798**	**373,171**	256,793	256,404
FHLB Advances (Note 11)	**157,000**	**156,851**	150,000	152,675
Junior Subordinated Obligations Issued to Unconsolidated Subsidiary Trusts (Note 12)	**20,000**	**20,007**	20,000	20,651

NOTE 25: PARENT ONLY FINANCIAL INFORMATION (In Thousands)

Condensed financial information for Arrow Financial Corporation is as follows:

BALANCE SHEETS	**December 31,**	
ASSETS	**2005**	2004
Interest-Bearing Deposits with Subsidiary Banks	**$ 382**	$ 9,293
Securities Available-for-Sale	**491**	453
Investment in Subsidiaries at Equity	**139,121**	130,443
Other Assets	**3,921**	3,745
Total Assets	**$143,915**	$143,934
LIABILITIES		
Junior Subordinated Obligations Issued to Unconsolidated Subsidiary Trusts	**$ 20,000**	$ 20,000
Other Liabilities	**6,494**	5,900
Total Liabilities	**26,494**	25,900
SHAREHOLDERS' EQUITY		
Total Shareholders' Equity	**117,421**	118,034
Total Liabilities and Shareholders' Equity	**$143,915**	$143,934

STATEMENTS OF INCOME	**Years Ended December 31,**		
Income:	**2005**	2004	2003
Dividends from Bank Subsidiaries	**$ 6,100**	$ 6,975	$ 9,726
Interest and Dividends on Securities Available-for-Sale	**32**	41	7
Other Income (Including Management Fees)	**645**	758	753
Net Gains on the Sale of Securities Available-for-Sale	**45**	11	13
Total Income	**6,822**	7,785	10,499
Expense:			
Interest Expense	**1,270**	1,208	842
Salaries and Benefits	**219**	202	199
Occupancy and Equipment	**3**	4	1
Other Expense	**632**	763	669
Total Expense	**2,124**	2,177	1,711
Income Before Income Tax Benefit and Equity in Undistributed Net Income of Subsidiaries	**4,698**	5,608	8,788
Income Tax Benefit	**746**	579	410
Income Before Equity in Undistributed Net Income of Subsidiaries	**5,444**	6,187	9,198
Equity in Undistributed Net Income of Subsidiaries	**13,195**	13,291	9,719
Net Income	**$18,639**	$19,478	$18,917

NOTE 25: PARENT ONLY FINANCIAL INFORMATION (Continued)

STATEMENTS OF CASH FLOWS	Years Ended December 31,		
	2005	2004	2003
Operating Activities:			
Net Income	**$18,639**	$19,478	$18,917
Adjustments to Reconcile Net Income to Net Cash Provided by Operating Activities:			
Undistributed Net Income of Subsidiaries	**(13,195)**	(13,291)	(9,719)
Net Gains on the Sale of Securities Available-for-Sale	**(45)**	(11)	(13)
Shares Issued Under the Directors' Stock Plan	**120**	75	57
Compensation Expense for Allocated ESOP Shares	**178**	405	91
Changes in Other Assets and Other Liabilities	**633**	497	(1,328)
Net Cash Provided by Operating Activities	**6,330**	7,153	8,005
Investing Activities:			
Proceeds from the Sale of Securities Available-for-Sale	**469**	160	331
Purchases of Securities Available-for-Sale	**(512)**	(202)	(346)
Net Cash Used in Investing Activities	**(43)**	(42)	(15)
Financing Activities:			
Exercise of Stock Options and Shares Issued to the Employees' Stock Purchase Plan	**1,204**	1,591	1,241
Proceeds from Issuance of Trust Preferred Securities	**---**	10,000	10,000
Repayment of Trust Preferred Security	**---**	(5,000)	---
Tax Benefit for Disposition of Stock Options	**684**	409	109
Purchase of Treasury Stock	**(7,528)**	(2,453)	(5,557)
Cash Dividends Paid	**(9,558)**	(9,000)	(8,225)
Net Cash Used in Financing Activities	**(15,198)**	(4,453)	(2,432)
Net Increase in Cash and Cash Equivalents	**(8,911)**	2,658	5,558
Cash and Cash Equivalents at Beginning of the Year	**9,293**	6,635	1,077
Cash and Cash Equivalents at End of the Year	**$ 382**	$ 9,293	$ 6,635
Supplemental Disclosures to Statements of Cash Flow Information:			
Interest Paid	**$ 1,270**	$ 1,208	$ 842
Impact of Deconsolidation of Subsidiary Trusts upon Adoption of FIN 46R as Described in Note 1:			
Increase in Junior Subordinated Obligations	**---**	---	15,000
Decrease in Trust Preferred Securities	**---**	---	(15,000)
Increase in Other Assets Representing Investment in Unconsolidated Trusts	**---**	---	(465)

NOTE 26: CONCENTRATIONS OF CREDIT RISK

Most of Arrow's loans are with customers in northern New York. Although the loan portfolios of the subsidiary banks are well diversified, tourism has a substantial impact on the northern New York economy. The commitments to extend credit are fairly consistent with the distribution of loans presented in Note 4. Generally, the loans are secured by assets and are expected to be repaid from cash flow or the sale of selected assets of the borrowers. Arrow evaluates each customer's creditworthiness on a case-by-case basis. The amount of collateral obtained, if deemed necessary by Arrow upon extension of credit, is based upon management's credit evaluation of the counterparty. The nature of the collateral varies with the type of loan and may include: residential real estate, cash and securities, inventory, accounts receivable, property, plant and equipment, income producing commercial properties and automobiles.

SUMMARY OF QUARTERLY FINANCIAL DATA (Unaudited)

The following quarterly financial information for 2005 and 2004 is unaudited, but, in the opinion of management, fairly presents the results of Arrow. Earnings per share amounts have been adjusted for the 2005 3% stock dividend.

SELECTED QUARTERLY FINANCIAL DATA
(In Thousands, Except Per Share Amounts)

	2005			
	First Quarter	**Second Quarter**	**Third Quarter**	**Fourth Quarter**
Total Interest and Dividend Income	**$16,867**	**$17,776**	**$18,294**	**$19,190**
Net Interest Income	**11,804**	**12,155**	**12,136**	**11,918**
Provision for Loan Losses	**232**	**176**	**218**	**404**
Net Securities Gains	**64**	**125**	**151**	**24**
Income Before Provision for Income Taxes	**6,381**	**6,686**	**6,999**	**6,676**
Net Income	**4,430**	**4,680**	**4,839**	**4,690**
Basic Earnings Per Common Share	**.42**	**.45**	**.47**	**.45**
Diluted Earnings Per Common Share	**.41**	**.44**	**.46**	**.45**

	2004			
	First Quarter	Second Quarter	Third Quarter	Fourth Quarter
Total Interest and Dividend Income	$17,302	$17,062	$17,038	$17,041
Net Interest Income	12,304	12,111	12,502	12,320
Provision for Loan Losses	285	254	205	276
Net Securities Gains (Losses)	210	---	(9)	161
Income Before Provision for Income Taxes	7,117	6,819	7,273	7,229
Net Income	4,865	4,698	4,968	4,948
Basic Earnings Per Common Share	.47	.45	.48	.47
Diluted Earnings Per Common Share	.46	.44	.47	.46

a. Clarification
The Consultant shall interpret Contract Documents, provide clarifications, and make recommendations, by drawing and in writing, as required by DDC.

b. Prepare Supplementary Drawings
The Consultant shall promptly prepare any supplementary drawings that may be necessary for clarifying the contract documents.

c. Sealed and Signed
Supplementary drawings are to be sealed and signed by the Consultant or the Sub-consultant as appropriate.

d. Obtain Required Approvals
The Consultant shall obtain any approvals for supplementary drawings as necessary from applicable regulatory agencies and utilities.

7. Review of Contractor Coordination Documents

a. The Consultant shall review the Contractor's coordination documents and promptly report in writing to the DDC Resident Engineer on issues relating to meeting the project schedule and achieving the quality of work specified in the Contract Documents.

b. The Consultant shall systematically monitor the progress of all construction work scheduled and promptly report to DDC any conditions that may cause delays in the completion of the work.

8. Resolution of Design Errors or Omissions

a. The Consultant shall promptly submit to DDC any necessary correspondence, supplementary or revised drawings, specifications, negotiated cost estimates and any other documentation or coordination material to resolve design errors or omissions.
b. Upon approval of the required changes in the contract documents by DDC, the Consultant shall promptly provide to the Contractors all the documentation necessary to execute the work as revised.

9. Change Orders

a. Compensation for Change Orders
Payment shall be made for the Consultant to prepare additional drawings or otherwise modify the Construction Documents due to Owner directed Change Orders during the construction period resulting from scope changes, administrative changes and field conditions that could not reasonably have been anticipated prior to the time of bids, and which require design modifications.

b. Staffing Plan and Cost Proposal
Compensation shall be in accordance with the "Agreement," with the total amount shown on the staffing chart to constitute the maximum payable for the change order work. Within fourteen consecutive calendar days of any change order initiation, the Consultant shall

SOCIETE GENERALE, as Syndication Agent
and as a Lender.

By: /s/ Anne-Marie Dumortier
Director

KEYBANK NATIONAL ASSOCIATION, as
Documentation Agent and as a Lender

By: /s/ Mary K. Young
Senior Vice President

FIFTH THIRD BANK

By: /s/ Roy C. Lanctot
Vice President

NATIONAL CITY BANK

By: /s/ Marguerite C. Burtzlaff
Senior Vice President

NATIONAL CITY BANK, CANADA BRANCH, as the Lending Installation designated by National City Bank to make Non-Pro Rata Foreign Currency Loans to the Canadian Borrower on its behalf.

By: /s/ J. Andrew Riddell
Senior Vice President

3. Exhibits:

The following exhibits are incorporated by reference herein.

Exhibit Number	**Exhibit**
3.(i)	Certificate of Incorporation of the Registrant, as amended, incorporated herein by reference from the Registrant's Annual Report on Form 10-K for the year ended December 31, 1990, Exhibit 3.(a).
4.1	Shareholder Protection Rights Agreement dated as of May 1, 1997, between Arrow Financial Corporation and Glens Falls National Bank and Trust Company, as Rights Agent, incorporated herein by reference from the Registrant's Statement on Form 8-A, dated May 16, 1997, Exhibit 4.
4.2	Amended and Restated Declaration of the Trust by and among U.S. Bank National Association, as Institutional Trustee, Arrow Financial Corporation, as Sponsor and certain Administrators named therein, dated as of July 23, 2003, relating to Arrow Capital Statutory Trust II, incorporated herein by reference from the Registrant's Quarterly Report on Form 10-Q for the quarter ended September 30, 2003, Exhibit 4.1.
4.3	Indenture between Arrow Financial Corporation, as Issuer, and U.S. Bank National Association, as Trustee, dated as of July 23, 2003, incorporated herein by reference from the Registrant's Quarterly Report on Form 10-Q for the quarter ended September 30, 2003, Exhibit 4.2.
4.4	Placement Agreement by and among Arrow Financial Corporation, Arrow Capital Statutory Trust II and SunTrust Capital Markets, Inc., dated July 23, 2003, incorporated herein by reference from the Registrant's Quarterly Report on Form 10-Q for the quarter ended September 30, 2003, Exhibit 4.3.
4.5	Guarantee Agreement by and between Arrow Financial Corporation and U.S. Bank National Association, dated as of July 23, 2003, incorporated herein by reference from the Registrant's Quarterly Report on Form 10-Q for the quarter ended September 30, 2003, Exhibit 4.4.
4.6	Amended and Restated Trust Agreement by and among Wilmington Trust Company, as Institutional Trustee, Arrow Financial Corporation, as Sponsor and certain Administrators named therein, dated as of December 28, 2004, relating to Arrow Capital Statutory Trust III, incorporated herein by reference from the Registrant's Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 4.6.
4.7	Junior Subordinated Indenture between Arrow Financial Corporation, as Issuer, and Wilmington Trust Company, as Trustee, dated as of December 28, 2004, incorporated herein by reference from the Registrant's Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 4.7.
4.8	Placement Agreement by and among Arrow Financial Corporation, Arrow Capital Statutory Trust III and SunTrust Capital Markets, Inc., dated December 28, 2004, incorporated herein by reference from the Registrant's Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 4.8.
4.9	Guarantee Agreement by and between Arrow Financial Corporation and Wilmington Trust Company, dated as of December 28, 2004, incorporated herein by reference from the Registrant's Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 4.9.
10.1	Select Executive Retirement Plan of the Registrant effective January 1, 1992 incorporated herein by reference from Registrant's Annual Report on Form 10-K for the year ended December 31, 1992, Exhibit 10(m). *
10.2	1993 Long Term Incentive Plan of the Registrant, incorporated herein by reference from Registrant's 1933 Act Registration Statement on Form S-8, Exhibit 4.1 (File number 33-66192; filed July 19, 1993). *
10.3	1998 Long Term Incentive Plan of the Registrant, incorporated herein by reference from Registrant's 1933 Act Registration Statement on Form S-8, Exhibit 4.1 (File number 333-62719; filed September 2, 1998). *
10.4	Directors Deferred Compensation Plan of Registrant, incorporated herein by reference from Registrant's Annual Report on Form 10-K for the year ended December 31, 1993, Exhibit 10(n).*
10.5	Senior Officers Deferred Compensation Plan of the Registrant, incorporated herein by reference from Registrant's Annual Report on Form 10-K for the year ended December 31, 1993, Exhibit 10(o).*
10.6	Directors Stock Plan of the Registrant, as amended, incorporated herein by reference from Registrant's 1933 Act Registration Statement on Form S-8 (file number 333-110445, filed November 13, 2003).*

Exhibits incorporated by reference, continued:

Exhibit Number	**Exhibit**
10.7	2000 Employee Stock Purchase Plan of the Registrant, incorporated herein by reference from Registrant's 1933 Act Registration Statement on Form S-3 (File number 333-47912; filed on October 11, 2000).*
10.8	Award under Schedule A of Select Executive Retirement Plan to Thomas L. Hoy, dated May 2, 2001, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2001, Exhibit 10.15.*
10.9	Award under Schedule A of Select Executive Retirement Plan to John J. Murphy, dated May 2, 2001, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2001, Exhibit 10.16.*
10.10	Prototype of a change of control agreement between the Registrant and certain officers (excluding senior officers) of the Registrant or its subsidiaries, as entered into from time to time, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2003, Exhibit 10.11.*
10.11	Agreement and Plan of Reorganization by and among Glens Falls National Bank and Trust Company, Arrow Financial Corporation, 429 Saratoga Road Corporation, Capital Financial Group, Inc. and John Weber dated November 22, 2004, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 10.14.
10.12	Post-Closing Payment Agreement by and among Glens Falls National Bank and Trust Company, Arrow Financial Corporation, 429 Saratoga Road Corporation, Capital Financial Group, Inc. and John Weber dated November 22, 2004, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 10.15.
10.13	Employment Agreement between Arrow Financial Corporation, Glens Falls national Bank and Trust Company, Capital Financial Group, Inc. and John Weber dated November 29, 2004, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2004, Exhibit 10.16.*
14	Financial Code of Ethics, incorporated herein by reference from the Registrant’s Annual Report on Form 10-K for the year ended December 31, 2003, Exhibit 14.

* Management contracts or compensation plans required to be filed as an exhibit.

The following exhibits were submitted with the original filing:

Exhibit Number	Exhibit
3.(ii)	By-laws of the Registrant, as amended.
10.14	Employment Agreement among the Registrant, its subsidiary bank, Glens Falls National Bank and Trust Company, and Thomas L. Hoy dated January 1, 2006.*
10.15	Employment Agreement among the Registrant, its subsidiary bank, Glens Falls National Bank and Trust Company and John J. Murphy dated January 1, 2006.*
21	Subsidiaries of Arrow

* Management contracts or compensation plans required to be filed as an exhibit.

The following exhibits are submitted herewith:

Exhibit Number	Exhibit
23	Consent of Independent Registered Public Accounting Firm
31.1	Certification of Chief Executive Officer under SEC Rule 13a-14(a)/15d-14(a)
31.2	Certification of Chief Financial Officer under SEC Rule 13a-14(a)/15d-14(a)
32	Certification of Chief Executive Officer under 18 U.S.C. Section 1350 and Certification of Chief Financial Officer under 18 U.S.C. Section 1350

KEYBANK NATIONAL ASSOCIATION, as
Documentation Agent and as a Lender

By: /s/ Mary K. Young
Senior Vice President

FIFTH THIRD BANK

By: /s/ Roy C. Lanctot
Vice President

NATIONAL CITY BANK

By: /s/ Marguerite C. Burtzlaff
Senior Vice President

NATIONAL CITY BANK, CANADA BRANCH, as the Lending Installation designated by National City Bank to make Non-Pro Rata Foreign Currency Loans to the Canadian Borrower on its behalf.

By: /s/ J. Andrew Riddell
Senior Vice President

List of Approved Key Personnel for N66604-26-R-0163

The following Key Personnel are approved for this task order. Any changes to the below list require approval in accordance with clause C-237-H002 “SUBSTITUTION OF KEY PERSONNEL”.

Name	Labor Category	Award or Modification
Insert the name in the following format: Last Name, First Name	Insert the labor category	Insert either “Award” if approved at award or the modification number that the key person was

Organizational Structure

Aligning funding and staff with clearly defined roles and responsibilities makes it easier to agree on priorities and implement projects. Mapping and documenting processes can help establish more transparent norms and practices, reducing redundancies and improving communication and handoffs between teams to prevent project development oversights that can later require change orders to rectify.

See Process best practice strategy for more information.

Putting the Project First

Clear priorities and shared goals between project parties are essential for successful projects. A project-first culture that prioritizes project quality empowers project staff to dedicate sufficient time to develop realistic project deliverables and incorporate stakeholder feedback. Similarly, projects benefit when each involved party understands and appreciates the others' goals early in project development. This helps to prevent rushing projects to move through programmed funds, which can lead to downstream cost and schedule consequences.

3.8 Chapter Summary

There are many approaches to preventing and minimizing the likelihood and impact of construction change orders. The best practice strategies presented in this chapter can be used independently or in combination with each other to address technical, organizational, financial and funding causes of change orders depending on the pervasiveness of change orders in a program. Practitioners can consider how to:

- Implement formal processes that ensure quality, consistency, and completeness of technical work products
- Use risk management to improve decision-making that targets project elements that can potentially cause change orders
- Promote early and frequent communication with internal and external stakeholders to fortify project teams and facilitate information exchange as well as the understanding of stakeholder roles and responsibilities in the project delivery lifecycle to ensure project designs and agreements are completed with input from all perspectives
- Employ alternative contracting methods where appropriate to balance project sponsor and contractor risks and minimize project vulnerabilities that can lead to change orders
- Access a robust and skilled workforce that has the capacity to manage projects and produce work free from errors, omissions, and oversights that require change orders to correct

Contract Submissions

- There are several ways for agency personnel to submit contract actions to the Comptroller for registration.*
    1. Physical Hand-Delivery
    2. Electronic Submissions (PASSPort, Digital Interfaces, etc.)
- ***Note**: Agencies may also need to file packages associated with self-registered contracts (e.g. M/WBE NCSP Contracts).

Physical Hand-Delivery

Hand-delivered submissions can be made to the Central Imaging Facility (CIF):

- **Address**: 1 Centre, Street, Room No. 1225, New York, NY 10007
- **Phone**: (212) 669-3741
- **Fax**: (212) 815-8780

CIF Hours: Monday-Friday, 9am-5pm (closed on City holidays).

Electronic Submissions

- **Hours**: 9am-5pm (for same day processing)
- **Reminders:**
  - Electronic submissions after these hours will not be considered received until the following business day for the purposes of the Comptroller's 30-day review.
  - Electronic submissions made via PASSPort or a similar system will be validated to confirm that approved procurement methods were selected, and that documents corresponding to mandatory fields were attached.
  - If your submission is rejected due to a validation issue, follow error messages guidance or contact the Comptroller's Bureau of Information Systems and Technology (BIST) at helpdesk@comptroller.nyc.gov.

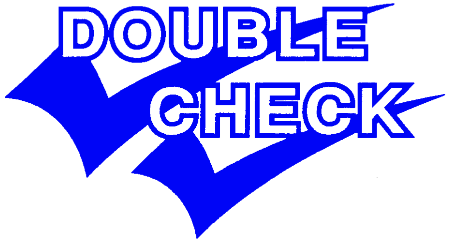
Important Last-Minute Checks

Mandatory Documents

- Each Contract Package must include the following documents:
  - Completed CIF Cover Sheet
  - Final Advice of Award
  - All other appropriate and required documentation pursuant to PPB Section 2-12, or other applicable guidelines.
- For modifications, use the New Document Name Field or the Modification Tab in FMS to enter details about the change.
- Note: Non-procurement actions (such as Real Property) or revenue contracts have different documentation requirements. Please reach out to the Bureau of Contract Administration if you have any questions.

Final Checks Before Submission

- Double check the following before submitting your contract to avoid unnecessary registration delays:
  - Ensure that award method codes, contract type codes, funding codes, and contract terms are correct.
  - Contract/Version #s, OCA #s, Vendor Name/Ids, EPIN #s, and insurance information should be consistent across the Cover Sheet, the Advice of Award, and other contract documents.
  - **Electronic Submissions**: Attachments should be properly labeled and filed under the correct document categories.



FMS Codes

FMS Data Codes and Descriptions

The following identifiers provide important details about a contract and how it was solicited:

- Contract “Award Method” Codes
- Contract “Type” Codes
- Contract “Category” Codes
- Contract “Award Level” Codes

Note: Inputting inaccurate or erroneous codes may result in registration delays.

FMS Award Methods

- "Award Method" indicates the type of procurement used in selecting the vendor.
- Agencies may only have permission to use certain Award Methods.
- Consult PPB rules or related guidance to verify selected code matches sourcing method.
- FMS Award Methods

FMS Contract Type

- "Contract Type" indicates the purpose or nature of a contract.
- Agencies are required to identify the contract type in order to enable better tracking of expenses and revenues.
- FMS Contract Types

FMS Category Codes

- Contract “Category” Codes indicates the classification of the contract .
- These identifiers provide more detailed information about a contract’s function.
- FMS Award Categories

FMS Award Level Codes

- "Award Level" codes provide additional detail about the basis of the award (e.g. Award Level 7 used on an 01, Competitive Sealed Bid, would indicate that it was a Best Value Bid)
- The city uses 8 Award Level Codes
- FMS Award Levels

Agency Request Portal

Agency Request Portal

- The Agency Request Portal (ARP) allows agency staff to request assistance from the Comptroller to correct errors with a pending contract, or to modify details in a registered contract.
- Such requests will only be accepted via the ARP (rather than phone or email).
- **Note**: Agency staff are encouraged to double check critical contract and vendor details before submission in order to minimize the need for corrections via the ARP.

Unlocks and Overrides

- Agency staff can use the ARP to make **unlock** or **override** requests.
- **Unlock Requests:**
  - Enables agencies to modify details such as: contract amount, award method, contract type, accounting lines, etc.
  - Can only be submitted for actions that are pending registration with the Comptroller.
- **Override Requests:**
  - Allows agencies to modify details for a registered contract.
  - A corresponding CTR or MAR document must be created in FMS reflecting the requested change before an override request is submitted.
- **Note**: Changes to a vendor details (e.g. name or address) submitted via the ARP may require additional approval by the Comptroller's Vendor Validation Unit.

Accessing and Navigating ARP

Link to ARP: [Agency Request Portal (comptrollernyc.com)]

- Complete the email validation process (each user should only be prompted their first time using the ARP).
- Complete all fields in the form.
- Use the justification field to provide important details about your request.
  - **Note:** changes to the award method or fund code may impact the review process. It is critical that such requests be enumerated clearly in the justification section in order to avoid significant registration delays.
  - Users will receive an email with a confirmation number after submitting a request.
- Use the **Check Status** fields on the home screen to review submitted requests.
  - Users can also cancel a pending request from the check status screen.
  - Users will be automatically emailed when there is a status update.

Purchase Orders: Purchase orders are effective and binding upon the contractor when placed in the mail.

Quantities: The quantities listed are estimated only and based upon the filed requirements. However, vendor payments will be based on quantities actually ordered and received during the contract period.

Minimum Order: No minimum order.

Contract Period: It is the intention of the State to enter into a contract for a term of (initial contract term), as indicated on the IFBs. Agency reserves the right to extend this contract under the same terms and conditions for an additional period of (allowed in IFB but not to exceed a total contract term of 5 years).

Price Adjustment: The contractor may request a price change for any extension period(s) if allowed and limited to the price adjustment clause(s) per the IFB. Documentation of the increases must accompany the request. Price reductions may be offered at any time during the term of the contract including renewal periods.

Appendix A: The contractor agrees to comply with Appendix A, New York State Standard Contract Clauses.

Terms and Conditions: The contractor agrees to comply with all the specifications of (agency) Bid Number (x) dated (specific date).

Notes to Agency as applicable:

- Purchase orders should reference this contract number (PAxx)
- When placing orders against this contract, please forward a copy of the P.O. to: (Agency main office, address, phone number, fax number)
- Agencies are urged to process vouchers expeditiously.

(see item list on next page)

CONTRACT NO. PAxx - Public Safety Officers Uniforms (SAMPLE ONLY)

Item Number 1:

Estimated Quantity: 1005

Name and number of material proposed to be furnished: Raeford 571-34453

Style number or catalog reference of finished garment: Special

Manufacturer: Clifton Shirt

Delivery: 45 days A/R/O (after start up)

Item: Long Sleeve Shirts (men)
with embroidered insignia patch, as per specification
Color: Blue/Gray
Sizes:

Neck 14 C 17 2, Sleeve 31-35 **Unit Price:** 42.70

Neck 18 C 20 2, Sleeve 36-38, or long body **Unit Price:** 46.70

Item Number 2:

Estimated Quantity: 125

Name and number of material proposed to be furnished: Raeford 571-34453

Style number or catalogue reference of finished garment: Special

Manufacturer: Clifton Shirt

Delivery: 45 days A/R/O (after start up)

Item: Long Sleeve Shirts (women)
with embroidered insignia patch, as per specification
Color: Blue/Gray

Sizes:

S,M,L **Unit Price:** 42.70

XL **Unit Price:** 46.70

which procurement needs to contribute. However, the development toward strategic, smart, value-driven, and sustainable procurement implies that the societal importance of procurement practices is becoming more prominent. This requires a rebalancing of the disciplinary perspectives.

For example, the legal perspective should rebalance its priorities and move from a ‘cannot’ mentality to a ‘can do’ mentality where not necessarily the legally safest option is advised, but the safest one that creates the most public value. Price considerations will always remain important in purchasing decisions (taxpayers’ money should not be squandered), however, the focus should move toward getting the most value for your money, rather than spending the least. This entails a shift from a focus on purchase price to a life cycle perspective on quality and capturing the most public value from a transaction.

Societal challenges have a lot to do with how politicians make decisions and how to act upon them. In most European countries, politicians represent the democratic values and expectations of the people. The political system has been driving the policy agenda as well as financial frameworks and resulting budgets. To be able to move toward a more innovative and circular way of procuring with stable supplier relations, a forward-looking and less political way of governing seems necessary. One that is less focused on the short term and election cycles, but more visionary and programmatic with long-term partnerships, and an emphasis on human well-being and value co-creation rather than economic prosperity.

8.4 Conclusion: Become a Change Agent

The developments in the field of public procurement and subsequent implications for public procurement practices and organizations suggest major changes in our procurement systems. Truly bringing public procurement forward into a new era requires change agents to drive these changes. A change agent can be anybody, an individual or a team, from inside or outside the organization, that takes responsibility for initiating, sponsoring, directing, managing, or implementing a change (Caldwell, 2003). This responsibility is something that does not have to be imposed on the change agent (it not necessarily part of the job) but is often a task that change agents take up, out of a desire to do something and make change happen (Grandia, 2015). If one thinks of change agents, the image of a top or senior manager might also spring to mind. However, studies show that anybody at any level can become a change agent, from interns to director generals. It merely requires a person to act (Caldwell, 2003; Kendra & Taplin, 2004).

There are many actions that change agents can carry out to effect change, such as envisioning, initiating, sponsoring, adapting, or carrying forward change. One could also build support, provide advice, expertise, or process skills, or contribute by interviewing, directing, managing, speaking, or presenting. Furthermore, listening, reflecting, writing, cooperating, refining, giving feedback, and/or training or educating are important activities that change agents perform. The mentioned actions remain rather vague, as there is no universal change agent model that shows what kind of actions are required in which situations and under which circumstances (Caldwell, 2003).

Becoming a successful change agent, however, does not necessarily require enormous actions or plans, even the smallest initiatives can ignite a change or offer a breakthrough. Sometimes merely asking questions about why things go a certain way can, in the end, lead to major changes. Long-term gradual accumulation of many small changes has been found to successfully lead to large changes in the end.

We therefore challenge you, the reader, to become an agent of change and help drive public procurement forward into the new era. As an agent of change you could, for example, present new ideas to create more public value with procurement (initiate), talk enthusiastically about the possibilities and necessity of the change (motivate), share knowledge about developments or new alternatives (educate), advise on how to incorporate new knowledge (advise), or arrange the necessary tools to make it happen (solve problems). All these types of actions can help make other stakeholders in the procurement process more willing and able to make changes and use the potential of public procurement for achieving societal impact. Let’s be this change together.

8.5 Summary

This chapter first shortly summarized the topics discussed in the book, followed by a discussion of the four main trends and developments in the field of public procurement that can be identified. First, a move from efficiency and cost-based thinking toward value-driven and sustainable oriented procurement processes could be observed. Second, a move from a focus on administrative operational thinking toward digitalized and smart decision-making in procurement processes seems visible. Third, procurement appears to move from a front-end practical purchasing instrumental perspective to procurement as a strategic life cycle engagement process. And fourth, procurement moves from supporting the formal dyadic supply chain relations to public procurement to facilitating relational networks and ecosystems. This chapter then explained that the trend toward more value-driven, smart, life cycle-oriented, and relational ecosystem procurement processes has implications for the public procurement practices of the future. The following four main implications or challenges for public procurement that need to be addressed to bring public procurement forward were identified: (1) more flexible and adaptive ways of governing the relationship between buyer and supplier, (2) integration of public procurement values in all parts of the public organization, (3) a need for public procurers with different capabilities and competences, and (4) rebalancing the multiple perspectives on public procurement. The final chapter of this book finished with an explanation of why it is important that change agents step up and help public procurement move into a new era and challenge the reader to become such a change agent themselves.

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J. Grandia, L. Volker (eds.), *Public Procurement*,
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3. Contract Development Unit (CDU)

The DYCD **Centralized Contracting Unit (CDU)** offers specialized contract support and guidance to providers funded under all DYCD Initiatives. CDU works directly with PACE (our Procurement Department) and DYCD funded providers on the development of their contracts from initiation to submission, by moving them through the City's PASSPort Portal system to ensure timely registration. CDU's role in the contract management process starts once the contract is initiated in PASSPort. CDU Contract Associates will review documents for accuracy and completion and track the contract until it is registered. Contract Associates are also available to assist with troubleshooting problems that may arise with PASSPort.

SECTION ONE: THE BUDGET

BUDGET OVERVIEW

After a contract is registered in PASSPort, Providers must prepare and submit a budget based on the proposed services, funding availability and contract term.

Below are the main budget item categories used by DYCD in PASSPort:

Passport Budget Categories		
Accounting Costs	Milestone*	Remediation*
Allowance	Operations and Support	Rent
Analysis*	Operations and Support: Client Stipend	Salaried Employees
Audit Expense	Operations and Support: Client Transportation	Scoping*
Build*	Operations and Support: Equipment	Study*
Construction*	Operations and Support: Incentive Payments/Bonus	Task Order*
Consultants	Other	Transportation
Cost-of-Living Adjustment	OTPS Contracted Services	Unallocated Funds
Deliverable*	OTPS Contracted Services: Sub-contractors	Utilities
Design*	Personal Service Fringe Benefits	Vendors
Equipment	Personnel Service Salary	Work Order*
Hourly Employees	Phase*	
Indirect*	Professional Services	
Indirect Rate	Program Income*	
Legal Costs	Rate*	
Categories **NOT** Utilized by DYCD*		

Completed budgets are submitted via PASSPort and first routed to the assigned DYCD Budget Compliance Associate. The budget will be reviewed by fiscal staff within DYCD who will either approve the Budget or return it to the Provider for revision.. The final budget approved by DYCD will be included and made a part of the Provider's contract.

GENERAL INFORMATION

Below is information to keep in mind as a DYCD budget is completed in PASSPort.

The first header in PASSPort is the Fiscal Year Budget Information which is pre-populated with the FY Start Date, FY End Date and the FY Budgeted Amount applicable to the contract.

Fiscal Year Budget Information

FY Start Date	FY End Date	FY Budgeted Amount	Invoiced Amount	Remaining Amount	Invoiced Value	Remaining Value	FY Encumbered Amount	FY Closed Amount	Remaining Encumbered Amount	FY Rolled Amount
7/1/2023	6/30/2024	0.00	0.00	0.00	0.00	0.00	9,550.00	0.00	9,550.00	0.00

1 Result(s)

Operating Period

The term of the contract (start date to end date) may overlap with more than one Fiscal Year. The City's Fiscal Year runs from July 1st to June 30th.

PASSPort

Providers will submit their budgets through PASSPort . If a contract spans multiple years, the next year's budget template becomes available for submission within the final quarter of the current fiscal year.

Insurance

Providers that choose not to buy into the CIP Insurance Program must provide DYCD with a Certificate of General Liability Insurance, as well as any renewal certificates required during the contract term. Providers are required to have General Liability Insurance in the sum of not less than one million dollars ($1,000,000) per occurrence to protect Providers themselves and the City of New York and its officials and employees against claims, losses, damages, etc. Required certificates not presented in a timely manner may result in a delay in contract registration or may result in suspension of a contract. The policy must include theft insurance to guard against loss of equipment because of a break-in or robbery. Each Provider must be covered for loss due to burglaries, vandalism, fire, or floods that affect equipment or furniture that is leased or purchased with DYCD funds. If such equipment is lost or stolen, the Provider must obtain a police report detailing the nature of the incident as well as submit a claim to the insurance carrier. In addition, the Provider must submit an official report to DYCD. The Provider must replace lost or stolen DYCD equipment with funds obtained from settlement of the claim. The Insurance Compliance Unit must receive written notification within **fifteen (15)** days if the policy is cancelled during the contract term.

Insurance Requirements as of FY2024

The New York City Comptroller's Office now requires that the Certificates of General Liability Insurance have the National Association of Insurance Commissioner (NAIC #) included on the right of the page, next to the insurer A box.

The City Law Department requires DYCD to ensure that all the Certificates of General Liability for our contracted Providers contain the following statement in the box labeled "Description of Operations/Locations/Vehicles":

"The City of New York, including its officials and employees, is included as Additional Insured."

Furthermore, Programs located in Department of Education (DOE), or New York City Housing Authority (NYCHA) facilities must carry insurance that covers and names the City of New York and DOE or NYCHA, as the case may be, as Additional Insured. The Certificate for such a program must contain the following statement in the box labeled "Description of Operations/Locations/Vehicles":

"The City of New York, and the Department of Education of the City School District of the City of New York [or New York City Housing Authority], including their officials and employees, are included as an Additional Insured."

Additionally, each certificate of insurance must be accompanied by a copy of the endorsement that is used for the Provider's policy. If the endorsement contains a box titled "Location(s) Of Covered Operations," then it must list the location where services are being provided.

If services are provided in multiple locations, under "Location(s) Of Covered Operations," the Provider must include the following language: "All locations of operations that are listed in the contract(s)" in lieu of having to list each location where services are provided.

Samples of the Certificates of Insurance and endorsements are available on the DYCD website.

Providers must make available a Certificate of Insurance, together with a notarized Broker's Certificate, and the Additional Insured Endorsement, to DYCD. Providers must also submit any renewal certificates required during the contract term. Required certificates not presented in a timely manner may result in suspension of the contract. DYCD retains the right to enroll a non-compliant Provider in CIP and to withhold 4.5% of the contract to cover the cost of CIP participation.

Additionally, Providers must also provide proof of Workers' Compensation, Disability Insurance and Professional Liability (if applicable), as well as any renewal certificates required during the contract term. Please note that the ACORD forms are not acceptable proof of Worker's Compensation and Disability Insurance. Acceptable forms include but are not limited to: C-105.2, U-26.3, SI-12, GSI-105.2, DB-120.1, DB-155 and CE-200.

Providers can email these proofs of insurance documents directly to DYCDInsurance@dycd.nyc.gov for review.

Automobile Liability Insurance

a. If vehicles are used in the provision of services under this Agreement, then the Contractor shall maintain Business Automobile Liability insurance in the amount of at least One Million Dollars ($1,000,000) for each accident combined single limit for liability arising out of ownership, maintenance, or use of any owned, non-owned, or hired vehicles to be used in connection with this Agreement. Coverage shall be at least as broad as the most recently issued ISO Form CA0001.

b. If vehicles are used for transporting hazardous materials, then the Business Automobile Liability Insurance shall be endorsed to provide pollution liability broadened coverage for covered vehicles (endorsement CA 99 48) as well as proof of MCS-90."

Please submit copies of the policy to DYCDInsurance@dycd.nyc.gov.

All other mandatory insurance policies must be made available for inspection by DYCD staff, CPA Auditors, and/or other authorized agents

Employers

Employer's FICA and MTA Tax are budgeted at **8.25%** of total salaries. The maximum of wages taxed for the Social Security portion of FICA can be found at www.ssa.gov. Please note that these rates and dollar amounts are determined by the Federal government and are subject to change.

The Metropolitan Commuter Transportation Mobility Tax (MCTMT) is imposed on employers and/or self- employed individuals who are required to withhold New York State income tax from wages and whose payroll expense for covered employees in the Metropolitan Commuter Transportation District (MCTD) exceeds **$312,500** in any calendar quarter. The MCTD consists of the five boroughs of New York City.

MTA Tax Rate Effective 07/01/2023: 0.60% (0.0060)
If a lower rate applies to your organization, then kindly provide documentation upon budget submission.

A full list of rates based on payroll expenses can be found at the following link:
https://www.tax.ny.gov/bus/mctmt/emp.htm

State Unemployment Insurance (SUI)

SUI is budgeted at the Provider's insurance rate for up to and including the "wage base," which is -- the amount of an employee's wages used to calculate an employer's Unemployment Insurance contributions. The table below lists the wage bases for 2017-2026.
Please note: Terminated staff as well as new staff hired within the same calendar year must be covered by SUI.

January 2017	$10,900	January 2024	$12,500
January 2018	$11,100	January 2025	$12,800
January 2019	$11,400	January 2026	$13,000
January 2020	$11,600		
January 2021	$11,800		
January 2022	$12,000		
January 2023	$12,300		

After 2026, the wage base will be adjusted on the first day of January each year to 16 percent of the state's average annual wage.

Medical Benefits, Life Insurance, Pension, Workers Compensation, and Disability costs are to be calculated based upon the Provider's policies.

COMPLETING THE DYCD BUDGET

Organizations can create and modify budgets of their active contracts in PASSPort, using Purchase Orders and Purchase Order Change Request (POCRs.)

The user creating and modifying budgets must have the ***Vendor Admin*** *or* ***Vendor Financial L2*** *roles in PASSPort.*

Below are the Item Categories applicable to DYCD contracts available in the detailed budget in PASSPort and a description of each category.

PASSPORT ITEM CATEGORIES

Accounting Costs

Expenses for financial record-keeping, reporting, and analysis. This includes hiring Accountants and Bookkeepers, accounting software subscriptions, and related expenditures.

Allowance

Health and Human Service contracts increased the contract authority by 25%; however, the full amount reflected in the allowance are not funds Providers immediately have access to. As additional funding is awarded to an existing contract and the contingency amendment is registered, DYCD will process a financial contract change request (FCCR) to the contract in PASSPort to reflect the additional funds. This will appear in PASSPort as a Change Order Budget. When submitting a Change Order Budget, please include the PDF emailed to your organization. Budgets will not be approved without the PDF included in the PASSPort budget submission.

Audit Expense

Expenses for financial review services, including fees for external auditors, internal audit teams, consulting, and related expenses

Consultants

A consultant hired on a health and human service contract is often a subject matter expert and does not perform or directly deliver a part of the prime contractor's programmatic contractual obligations. This means anyone assisting the Provider and not dealing directly with participants (ex: a STEM curriculum developer). Consultants cannot be salaried employees of the Provider. For each consultant listed, a signed Consultant Agreement must be uploaded to the Documents tab in the budget. Consultants retained by a Provider must enter into a written agreement detailing the specific tasks to be performed. Consultant Agreements and invoices must be maintained by the Provider for at least six (6) years. Consultant invoices must include the following details: rate, hours, type of services, date of service, consultant signature, and approval by the Provider's Executive Director or his/her designee.

Hourly Employees

For DYCD contracts, an Hourly Employee is defined as someone who is scheduled to work less than 35 hours per week and/or is paid on an hourly basis and retains a part-time employment status with the Provider. An hourly employee is allowed to work more than 35 hours per week during a specific season (e.g., summer) but must maintain an overall part-time status.

New York City Minimum Wage is $16/hr.

Indirect Rate

Indirect Rate (Indirect Costs)
Effective July 1, 2019, The City of New York Health and Human Services Cost Policies and Procedures Manual ("Cost Manual") governs the treatment and claiming of costs for health and human service contracts. If there is a conflict between the terms of DYCD's Fiscal Manuals and the Cost Manual, the Cost Manual shall take precedence. The Cost Manual was established to set guidance on indirect cost rate development and cost policies. The Cost Manual can be found at the link below:

For additional information on Indirect Costs, refer to the Nonprofit Resiliency Committee Indirect Implementation.
https://www1.nyc.gov/site/nonprofits/funded-providers/indirect-implementation.page

Indirect Rate is the only "Item Category" that is accepted for allocating your Indirect funds to your POCR.

Discretionary Contracts
The maximum Indirect Cost rate allowed by DYCD for Discretionary contracts is 10% of the cleared amount. City Council Discretionary contracts are exempt from the Cost Manual and from the ICR funding initiative.

Fiscal Agent Contracts
Providers under the Fiscal Agent with an Indirect Cost Rate must submit an attestation form for reimbursement.

Select the item category "Indirect Rate" when allocating indirect dollars to your PASSPort budget.

Legal Costs

Legal costs cover expenses for legal services vital to ensuring compliance, risk management, and protection of organizational interests. Funds cannot be used for litigation expenses, legal settlements, or legal judgements.

CURRENCY EQUIVALENTS

(Exchange Rate Effective May 31, 2026)

Currency Unit =	
122.75 BDT =	US$1

FISCAL YEAR

July 1 - June 30

Regional Vice President:	Johannes Zutt
Regional Practice Director:	Sebastian Eckardt
Country [Division] Director:	Jean Pesme
Practice Manager:	Hoon Sahib Soh, Tomas Ricardo Rosada Villamar
Task Team Leader(s):	Souleymane Coulibaly
Co- Team Leader(s):	Son Thanh Vo

****

NUWCDIVNPT VISITOR REQUEST INFORMATION NEEDED:

<u>Information must be on Company Letterhead**</u>**

1) **Company/Organization Name, Address, Telephone, and FAX Number**
2) **Cage Code**
3) **Visitor(s) Full Name (As it appears on government issued ID – i.e. Driver's License)**
4) **Visitor(s) Full Social Security Number**
5) **Visitor(s) Date of Birth**
6) **Visitor(s) Place of Birth (City & State)**
    a. If born outside the U.S, you MUST provide a Naturalization Number or US Passport
7) **Visitor(s) Citizenship**
8) **Level of Clearance:** UNCLASSIFIED or as appropriate
    a. SCI Level Access MUST be coordinated SSO to SSO
9) **Do they possess a CAC, Teslin or DBIDs Card? (Yes or No)**
    a. Please specify which visitor(s) have CAC, Retired Military/Dependent ID, DBIDS, or NOT APPLICABLE
10) **NUWCDIVNPT Point of Contact (Sponsor)**
    a. MUST be a NUWC Government Employee
11) **Specific Date(s) of Visit, to include Year**
12) **Purpose of Visit**
13) **Contract Number, if applicable**
    a. If NO Contract Number is provided, then the visit will be for 30 days at the UNCLASSIFIED Level
14) **Authorized Signature:**
    a. Name, Title, Signature, and Date of Requesting Company/Organization Authorizing Official (MUST be aUS Citizen)
    b. Smaller companies that do not have a Security Officer to sign may use another Official or Secretary.
    c. Individuals <u>MUST NOT</u> Sign their own request

****

Submit Visitor Request(s) and/or SECNAV form(s) 5512/1 via:

➢ **<u>JPAS SMO CODE</u>:** 666045 (If no access to JPAS, use methods below)

➢ **<u>EMAIL FORM TO</u>:** nuwc_npt_access_control@navy.mil

 <u>FAX FORM TO</u>: (401) 832-4396

ALL INCOMPLETE VISIT REQUESTS WILL BE DENIED

Please call NUWC visitor control at (401) 832-2152 if you want to verify any visits or status on vetting.

Critical Analysis on Government Contracts: An Introduction

Prof. (Dr.) K.V.S. Sarma
Vice-Chancellor, NLUJA, Assam

Government Contracts are nothing but legal agreements between two governments or between government and a private party. The government may be either central government or state government outlining the terms for provision of goods or services or with reference to immovable property also. Article 298 of Indian Constitution is dealing with Government Contracts. It must be in the format prescribed by Article 299 and in case the agreement is not in the prescribed format, it is not enforceable in the court of law. It stated that the agreement should be entered by the officer of the Government on behalf of Union or State Government and in the name of President of India or Governor of the State. Article 298 of the Indian Constitution of India confers the power to the Union as well as the State to carry out trade, or acquisitions, hold and dispose of property, and further make contracts for any purpose.

Article 299 prescribes the following conditions:

a) The concerned authority should enter the contract as an agent of the president of the union or by the governor of the State.

b) The contract must be in writing.

c) It must be entered on behalf of Union of India or on behalf of any State Government.

d) Agreements must be expressed in writing and executed by a person duly authorized by the President or the Governor on their behalf.

e) President or the Governor cannot be personally held liable for contract, it does not grant immunity to the government from the legal provisions of the contract.

It was held that a Government Contract has to comply with the provisions of Article 299 in addition to the requirements of Indian Contract Act, 1872. The contractual liability of the Government is the same as that of any individual under the ordinary law of the contract. (State of Bihar v. Majeed AIR 1954 SC 245).

When an agreement is not signed in the name of the Governor of a State and the name of the Governor is not mentioned in the deed, it was held that it was not valid and binding on the Government. (Mrs. Aliakutty Paul v. The State of Kerala and Ors AIR 1995 Ker 291).

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT

1. CONTRACT ID CODE

PAGE OF PAGES

2. AMENDMENT/MODIFICATION NUMBER

3. EFFECTIVE DATE

4. REQUISITION/PURCHASE REQUISITION NUMBER

5. PROJECT NUMBER *(If applicable)*

6. ISSUED BY CODE

7. ADMINISTERED BY *(If other than Item 6)* CODE

8. NAME AND ADDRESS OF CONTRACTOR *(Number, street, county, State and ZIP Code)*

(X)

☐ 9A. AMENDMENT OF SOLICITATION NUMBER

9B. DATED *(SEE ITEM 11)*

☐ 10A. MODIFICATION OF CONTRACT/ORDER NUMBER

10B. DATED *(SEE ITEM 13)*

CODE

FACILITY CODE

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

☐ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offers ☐ is extended. ☐ is not extended.

Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods:
(a) By completing items 8 and 15, and returning ____________ copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or electronic communication which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by letter or electronic communication, provided each letter or electronic communication makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA *(If required)*

13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NUMBER AS DESCRIBED IN ITEM 14.

CHECK ONE	
☐	A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: *(Specify authority)* THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NUMBER IN ITEM 10A.
☐	B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES *(such as changes in paying office, appropriation data, etc.)* SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(b).
☐	C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:
☐	D. OTHER *(Specify type of modification and authority)*

E. IMPORTANT: Contractor ☐ is not ☐ is required to sign this document and return __________ copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (*Organized by UCF section headings, including solicitation/contract subject matter where feasible.)*

Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER *(Type or print)*

16A. NAME AND TITLE OF CONTRACTING OFFICER *(Type or print)*

15B. CONTRACTOR/OFFEROR

(Signature of person authorized to sign)

15C. DATE SIGNED

16B. UNITED STATES OF AMERICA

(Signature of Contracting Officer)

16C. DATE SIGNED

Previous edition unusable

STANDARD FORM 30 (REV. 11/2016)
Prescribed by GSA FAR (48 CFR) 53.243

INSTRUCTIONS *(Back Page)*:

Instructions for items other than those that are self-explanatory, are as follows:

(a) Item 1 (Contract ID Code). Insert the contract type identification code that appears in the title block of the contract being modified.

(b) Item 3 (Effective date).

(1) For a solicitation amendment, change order, or administrative change, the effective date shall be the issue date of the amendment, change order, or administrative change.

(2) For a supplemental agreement, the effective date shall be the date agreed to by the contracting parties.

(3) For a modification issued as an initial or confirming notice of termination for the convenience of the Government, the effective date and the modification number of the confirming notice shall be the same as the effective date and modification number of the initial notice.

(4) For a modification converting a termination for default to a termination for the convenience of the Government, the effective date shall be the same as the effective date of the termination for default.

(5) For a modification confirming the contracting officer's determination of the amount due in settlement of a contract termination, the effective date shall be the same as the effective date of the initial decision.

(c) Item 6 (Issued By). Insert the name and address of the issuing office. If applicable, insert the appropriate issuing office code in the code block.

(d) Item 8 (Name and Address of Contractor). For modifications to a contract or order, enter the contractor's name, address, and code as shown in the original contract or order, unless changed by this or a previous modification.

(e) Items 9, (Amendment of Solicitation Number - Dated), and 10, (Modification of Contract/Order Number - Dated). Check the appropriate box and in the corresponding blanks insert the number and date of the original solicitation, contract, or order.

(f) Item 12 (Accounting and Appropriation Data). When appropriate, indicate the impact of the modification on each affected accounting classification by inserting one of the following entries:

(1) Accounting classification ....
Net increase $ ....

(2) Accounting classification ....
Net decrease $ ....

NOTE: If there are changes to multiple accounting classifications that cannot be placed in block 12, insert an asterisk and the words "See continuation sheet".

(g) Item 13. Check the appropriate box to indicate the type of modification. Insert in the corresponding blank the authority under which the modification is issued. Check whether or not contractor must sign this document. (See FAR 43.103.)

(h) Item 14 (Description of Amendment/Modification).

(1) Organize amendments or modifications under the appropriate Uniform Contract Format (UCF) section headings from the applicable solicitation or contract. The UCF table of contents, however, shall not be set forth in this document.

(2) Indicate the impact of the modification on the overall total contract price by inserting one of the following entries:

(i) Total contract price increased by $ ....

(ii) Total contract price decreased by $ ....

(iii) Total contract price unchanged.

(3) State reason for modification.

(4) When removing, reinstating, or adding funds, identify the contract items and accounting classifications.

(5) When the SF 30 is used to reflect a determination by the contracting officer of the amount due in settlement of a contract terminated for the convenience of the Government, the entry in Item 14 of the modification may be limited to --

(i) A reference to the letter determination; and

(ii) A statement of the net amount determined to be due in settlement of the contract.

(6) Include subject matter or short title of solicitation/contract where feasible.

(i) Item 16B. The contracting officer's signature is not required on solicitation amendments. The contracting officer's signature is normally affixed last on supplemental agreements.

a. *Community impacts:* temporary access disruptions, traffic interference, and nuisance during civil works;
b. *Construction-related impacts:* dust, noise, minor soil disturbance, and waste generation;
c. *Occupational risks:* worker safety during excavation and exposure to traffic and heat; and
d. *Social risks:* short-term business disruption, exclusion from digital services, insufficient and ineffective communication and stakeholder engagement, potential for sexual exposure and abuse/sexual harassment (SEA/SH) incidents, and potential dissatisfaction among BWA staff due to organizational reforms.

National Systems Assessment

7. Barbados has a comprehensive legal framework for environmental protection, occupational health and safety, land use, public health, and labor rights. Key national agencies include the Planning and Development Department (PDD), Environmental Protection Department (EPD), and Coastal Zone Management Unit (CZMU). Additionally, BWA's Water Resources and Environmental Management Unit (WREMU) oversees environmental management as well as providing support to occupational safety enforcement and customer service.

8. The assessment finds strong alignment between national systems and World Bank PforR Core Principles, particularly regarding environmental protection, worker safety, and grievance redress.

9. However, the assessment indicates that while there is alignment, BWA's institutional capacity and operational procedures may be insufficient to ensure consistent application of national environmental and social regulations because staffing, standardized Environmental Codes of Practice (ECoPs), standard operating procedures (SOPs), and monitoring systems seem limited.

10. Practical implementation requires reinforcement on:

a. Documented SOPs for environmental and social management and contractor oversight;
b. Construction-phase customer and public engagement including robust and monitorable grievance and incident reporting mechanisms; and
c. Standardized procedures for inclusive service access and stakeholder communication (e.g., documented offline/assisted options for users who can't use digital tools, plus clear, timely notifications and feedback channels during works).

Conclusions and Recommendations

11. The ESSA concludes that Barbados maintains a strong environmental and social regulatory framework, supported, among others, by the Planning and Development Act (2019, as amended 2020), the Environmental Impact Assessment (EIA) Regulations (2021), and sectoral instruments covering occupational safety, pollution control, and coastal management. These instruments collectively provide a sound legal foundation for managing environmental and social risks, define clear permitting procedures, and empower technical agencies—such as the Environmental Protection Department, Coastal Zone Management Unit, and Planning and Development Board—to impose mitigation and monitoring conditions as part of development approvals.

12. However, the assessment finds that institutional capacity and operational procedures within the BWA require strengthening to ensure consistent application of these safeguards at the program level. Legal provision implementation could be constrained by limited environmental and social staffing, the absence of standardized ECoPs, SOPs, and ad hoc monitoring and reporting systems.

13. Preliminary findings suggest that many of the required SOPs, templates, and field-level protocols are missing or not yet standardized. Confirmation of these elements will depend on additional inputs and documentation from BWA. Once this information is received, the ESSA will be updated to refine the analysis of institutional readiness.

14. Key recommendations to BWA include:

a. Strengthening BWA's environmental and social management procedures and staff capacity by clarifying institutional roles, allocating staff and resources for environmental and social oversight, and establishing internal procedures for supervision, monitoring, and reporting of environmental and social performance;
b. Formalizing screening and exclusion processes for ineligible activities;
c. Strengthening the current grievance mechanism to include construction-phased options for interaction with the public;
d. Integrating SEA/SH prevention and response protocols throughout operations;
e. Enhancing contractor OSH and environmental supervision; and
f. Ensuring accessibility of digital services and communication, particularly for vulnerable customers.

15. To assist in operationalizing these broader recommendations, the following four environmental and social items are included in the Program Action Plan (PAP):

a. Boosting BWA's environmental and social capacity by December 31, 2026;
b. Developing standard environmental and social protocols and SOPs prior to first tender or within six months of effectiveness, whichever comes first;
c. Launching a stakeholder engagement plan for wastewater reuse by December 31, 2026; and
d. Standardizing community engagement procedures to include grievance monitoring and redress for construction activities by December 31, 2026.

1. INTRODUCTION

16. Barbados has articulated a strategic plan to strengthen the reliability, efficiency, and resilience of its water and sanitation services. To support this effort, the World Bank is preparing a Program-for-Results (PforR) operation, the **Barbados Water Security and Sector Performance Program (P508674)** (the Program), to be implemented by the **Barbados Water Authority (BWA)**.

17. The Program will advance key priorities, including non-revenue water reduction, metering improvements, small-scale sewer pilots, and institutional strengthening such as digital customer service modernization.

18. As part of the due diligence required for PforR preparation, the World Bank conducts a suite of assessments covering technical, economic, fiduciary, climate/disaster resilience, and environmental and social aspects to confirm program feasibility, strengthen risk management, and align systems with World Bank Group Core Principles. Within this framework, this Environmental and Social Systems Assessment (ESSA) examines the national legal and institutional arrangements that govern the management of environmental and social effects, identifies strengths and gaps in existing systems, and recommends actions to enhance their capacity to address risks proportionate to the Program. The ESSA serves both as a diagnostic and as a practical tool to inform the Program design, including the Program Action Plan (PAP) and Disbursement Linked Indicators (DLIs).

19. As these challenges are not unique to Barbados, select additional countries in the Caribbean Region, and the World Bank have agreed to develop a regional approach to leverage synergies for overcoming these common barriers. Addressing the Region's complex water challenges requires highly specialized, continuous technical engagement, which most countries are not currently equipped to provide. Instead, they often rely on short-term external consultants or regional programs with limited scope, which leads to high costs, fragmented implementation, and limited continuity in reforms. This skills gap constrains countries' ability to develop strategic investments with an adequate level of readiness; to implement, operate and maintain critical infrastructure; and respond to climate-related pressures in a timely and sustainable manner. While several regional institutions provide support in water-related areas, regional gains of scale have been limited. Addressing this systemic constraint calls for a coordinated, wider-Caribbean approach to technical assistance, capacity building, and peer learning—shaped in the form of a regional water center of excellence. In this context, a Caribbean Regional Water Security Multi-Phased Approach (MPA) Program is proposed across several Caribbean States connected through an overarching regional platform and water security framework. The Barbados Program is the first activity of that MPA to move forward[1]. While this ESSA is stand-alone for the Barbados Program, the implementation of the program, including ESSA recommendations will benefit from Barbados's participation in the MPA.

A. Government Program

20. Barbados is among the most water-scarce countries in the world, with renewable freshwater resources estimated at only 304 m^3 per person per year, well below the international threshold for absolute scarcity. Nearly 90 percent of the national water supply depends on groundwater abstraction from fragile limestone aquifers, which are increasingly exposed to overextraction, saline intrusion, and contamination.

21. To address this structural vulnerability, the Government has developed the Barbados Water Authority Strategic Plan (SP) 2025–2029, which defines a comprehensive framework for building a resilient, efficient, and climate-smart water and sanitation sector.

[1] The first mover countries under this MPA are Barbados, St. Lucia, and Jamaica, with potential interest to be confirmed by Belize and Grenada; working together for overarching regional support with a regional Caribbean agency. Please see the World Bank Project PAD for the Program (P508674) for more details on the MPA. [INCLUDE LINK TO FINAL PAD IN PUBLISHED ESSA]

22. The Strategic Plan's overarching objective is to ensure equitable access to safe water and sanitation while preserving the island's scarce water resources. Its implementation is led by the BWA under the oversight of the Office of the Prime Minister.

23. The Strategic Plan combines investment in infrastructure with institutional modernization and policy reform. Its main lines of action include:

a. Reducing non-revenue water (NRW), which is currently around 55 percent, through improved asset management, leak detection, and digital monitoring systems;
b. Rehabilitating water supply infrastructure to enhance reliability and service quality;
c. Expanding sewerage coverage, which currently reaches only 5 percent of the population, using low-cost and decentralized sanitation technologies;
d. Improving septage collection, treatment, and reuse to protect groundwater resources; and
e. Upgrading of BWA's two wastewater treatment plants to tertiary level and reuse capacity.

24. However, BWA faces persistent financial and investment constraints that limit its ability to deliver these reforms. High operating expenses, which are mainly driven by personnel and energy costs for pumping and desalination, have eroded the utility's financial space. Between 2018 and 2024, BWA's annual revenues averaged US$86 million, enough to cover direct operating costs but insufficient to fund capital works. Average annual investment of US$15 million between 2021 and 2023 is vastly below the US$1 billion in sector investment needs estimated for the coming decades. At this rate, it would take more than 70 years to achieve current expansion objectives.

25. The Barbados Water Security and Sector Performance Program, with a total cost of about US$80 million, of which US$50 million is financed by International Bank for Reconstruction and Development (IBRD), is intended to help mitigate sector financing constraints and advance implementation of BWA's Strategic Plan (2025–2029) by channeling resources to infrastructure rehabilitation, digital modernization, and institutional strengthening.

B. ESSA Objective

26. The ESSA aims to evaluate the adequacy of Barbados' existing legal, regulatory, and institutional frameworks to manage the environmental and social risks and impacts associated with the Program. The ESSA identifies areas where national systems and BWA's practices are well aligned with the World Bank's PforR Core Principles, highlights potential gaps or capacity constraints, and recommends measures to strengthen risk management, enhance stakeholder engagement, and ensure that Program activities promote environmental sustainability, social inclusion, and protection of public and worker health and safety.

C. ESSA Methodology

27. This ESSA has been prepared in line with the World Bank's Program-for-Results Financing Environmental and Social Systems Assessment Guidance (2020). Successive drafts of ESSA were reviewed by BWA and stakeholders with consultations being held on December 11, 2025, and January 20, 2026. Comments and recommendations from those consultations are herein reflected. See Annex 1 for a summary of those consultations.

28. The methodology involved three main steps: (i) screening of potential environmental and social risks and impacts of Program-supported activities; (ii) assessment of the adequacy of Barbados' national systems and BWA's institutional arrangements to manage these risks in line with the Bank's Core Principles; and (iii) identification of opportunities for strengthening systems through targeted recommendations.

29. The assessment focused on identifying the environmental and social effects most likely to arise from Program activities. The identified effects were then organized into four categories: potential social benefits, potential social risks and impacts, potential environmental benefits, and potential environmental risks and impacts.

30. The assessment further examined the national policy, legal, and regulatory framework governing water and sanitation projects in Barbados, with attention to how these instruments apply to the types of works and impacts expected under the Program. Institutional roles and responsibilities of BWA and relevant oversight agencies were analyzed, alongside the established procedures for environmental screening, permitting, and compliance monitoring.

31. The review also considered BWA's track record in managing environmental and social aspects of both capital and maintenance works. Multiple sources of information were reviewed to inform the assessment, including:

a. Policies, technical guidelines, and procedures issued by BWA and the Planning and Development Department (PDD) on environmental and social management;
b. Environmental and social management plans, monitoring reports, and implementation records, including those prepared for internationally financed projects;
c. BWA's procedures for client communication and grievance handling; and
d. Additional documentation relevant to national environmental and social systems and BWA operations.

32. To complement the document review, a questionnaire was administered to BWA, and follow-up interviews were conducted with key personnel. These interviews, held remotely, provided further clarification on institutional practices, responsibilities, and implementation challenges.

33. The findings from this process were analyzed against the World Bank's Core Principles for PforR operations. The Core principles establish the minimum requirements a program must meet to ensure that environmental and social impacts are properly managed. They also aim to strengthen country systems and promote sustainable development outcomes.

a. **Core Principle #1:** Program environmental and social management systems are designed to (i) promote environmental and social sustainability in the Program design; (ii) avoid, minimize, or mitigate adverse impacts; and (iii) promote informed decision-making relating to a Program's environmental and social effects.

e. **Core Principle #2:** Program environmental and social management systems are designed to avoid, minimize, or mitigate adverse impacts on natural habitats and physical cultural resources resulting from the Program. Program activities that involve the significant conversion or degradation of critical natural habitats or critical physical cultural heritage are not eligible for PforR financing.

f. **Core Principle #3:** Program environmental and social management systems are designed to protect public and worker safety against the potential risks associated with (i) the construction and/or operation of facilities or other operational practices under the Program; (ii) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials under the Program; and (ii) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards.

g. **Core Principle #4:** Program environmental and social systems manage land acquisition and loss of access to natural resources in a way that avoids or minimizes displacement and assists affected people in improving, or at the minimum restoring, their livelihoods and living standards.

h. **Core Principle #5:** Program environmental and social systems give due consideration to the cultural appropriateness of, and equitable access to, Program benefits, giving special attention to the rights and interests of Indigenous Peoples/Sub-Saharan African Historical Underserved Traditional Local Communities, and to the needs or concerns of vulnerable groups.

i. **Core Principle #6:** Program environmental and social systems avoid exacerbating social conflict, especially in fragile states, post-conflict areas, or areas subject to territorial disputes.

34. The ESSA determines whether all core principles, or only a subset, are applicable to the Program. It then analyzes the system by comparing it against these principles to assess alignment and identify any gaps that may require additional measures.

35. Gaps between the current national and institutional systems and the Bank's requirements were identified. Where noted, recommendations were formulated to strengthen both system-level and project-level practices. These recommendations will be consolidated into the Program Action Plan (PAP), which specifies actions, responsibilities, and timeframes for improving the management of environmental and social aspects during Program implementation. Several of these actions will be supported through the Modernization Contract and can be reinforced by relevant DLIs.

2. PROGRAM DESCRIPTION

A. Program-for-Results

36. The Barbados Water Security and Sector Performance Program (P508674) will be implemented under the PforR financing instrument, with a total program cost of approximately US$80 million, of which US$50 million will be financed by IBRD.

37. The Program supports the implementation of the BWA Strategic Plan 2025–2029, focusing on strengthening the operational, institutional, and financial capacity of the BWA to enhance service delivery, improve water resources management, and expand safely managed sanitation.

38. The PforR will be structured around three results areas, each supported by a set of DLIs and Disbursement-Linked Results (DLRs) that aim to incentivize improved performance, institutional reforms, and financial sustainability.

39. **Results Area 1 – Building BWA capacity for efficient service delivery:** This area focuses on the modernization of BWA's operations and systems to improve service delivery, customer engagement, and financial and operational performance.

Table 1 – Disbursement Linked Indicators for Result Area 1

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 1: Improved BWA's services continuity capacity and reduced water losses	DLR 1.1 – BWA's disaster recovery and business continuity plan approved by BWA's Board DLR 1.2 – 24 priority District-Metered Areas (DMAs) operationalized DLR 1.3 – Eighty thousand (80,000) customer water meters installed and automated to a central BWA system country-wide DLR 1.4 – 24 DMAs reached 30% (or less) physical water losses
DLI 2: Improved BWA's customer responsiveness and financial performance	DLR 2.1 – Digital customer portal is integrated with commercial management systems through: contracting of a service provider for the design, establishment, and integration of said portal; and the installation and operationalization of said portal DLR 2.2 – 25% increase of customer complaints investigated within twenty (20) days of receipt DLR 2.3 – 20% increase in BWA's operating revenues

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 3: BWA's organizational reform, systems digitalization, and staff enhancement program are implemented	DLR 3.1 – Performance accountability mechanism for water supply and sanitation services established through: the signing of a performance and service level agreement between the Borrower and the PIE; and the publication of the annual performance report of the PIE's website. DLR 3.2 – Integrated utility-wide MIS is able to capture data and generate reports across three (3) priority modules (technical, financial, and WRM) DLR 3.3 – BWA's organizational and staffing reform implemented through: BWA's Board approval of the new BWA organizational structure; 30% of key staff positions filled; and 100 employees accredited

40. **Results Area 2 – Pollution management through country-wide safely managed sanitation services:** This area promotes the transition to safely managed sanitation across Barbados by strengthening wastewater and septage management systems and enabling water reuse. It will also support regulatory and operational improvements for septage collection, treatment, and reuse, as well as the development of low-cost sewerage pilots and stakeholder engagement programs to promote acceptance of wastewater reuse.

Table 2 – Disbursement Linked Indicator for Result Area 3

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 4: Reduced unsafe disposal of untreated wastewater and septage	DLR 4.1 – Water reuse regulation and an amendment to the septage management regulation are published in the official gazette DLR 4.2 – Two thousand (2,000) functional household sewerage connections established through lower-cost solutions DLR 4.3 – One hundred thirty-two thousand (132,000) KL of septage collected and treated

41. **Results Area 3 – Foundations for long-term water security:** This area strengthens Barbados's capacity to manage its limited water resources sustainably and adapt to climate variability. Activities will operationalize the Water and Wells Management Committee (WWMC) and enhance the Water Resources and Environmental Management Unit (WREMU), establish high-level coordination mechanisms across sectors, and improve data and monitoring systems.

Table 3 – Disbursement Linked Indicator for Result Area 3

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 5: Implemented institutional, legal and technical mechanisms for water security management and reporting	DLR 5.1 – Institutional and regulatory water security mechanisms are implemented through the Cabinet approval on the establishment of a high-level coordination mechanism on water security; and BWA's Board approval of the WWMC regulation[2]. DLR 5.2 – Country-wide groundwater monitoring and reporting system is implemented through an updated inventory of wells in Barbados; and public disclosure of annual water resource status based on WRIMS[3].

B. Institutional and Implementation Arrangements

42. BWA is a statutory corporation established under the Barbados Water Authority Act (1980, amended 2023), mandated to manage the national water supply and sewerage services that will serve as implementing agency for the Program. Its organizational structure includes technical and administrative departments such as Engineering and Planning, Operations, Quality Assurance, Human Resources, and Corporate Services.

43. BWA plans to establish a small Program Management Unit (PMU) to manage the overall administration of the Program, coordination across Departments, and coordination of the Modernization Contract implementation. BWA already has established a 'Transformation Committee' which comprises members of its board, directors and selected technical staff. BWA could utilize this same committee to ensure close coordination and synergized work between the Modernization Contract and BWA management and staff. Works are expected to be undertaken both directly by BWA staff and through private contractors, with BWA engineers and site supervisors anticipated to oversee compliance with technical specifications.

44. Regarding environmental and social responsibilities within BWA, the Water Resources and Environmental Management Unit (WREMU) holds the central mandate for water resources management, including the protection and sustainable use of groundwater, while occupational health and safety is managed by the Occupation Safety and Health (OSH) Committee and the OSH function within the Human Resources Department. Customer complaints and service-related grievances are handled through the Customer Service Division, with statutory oversight of tariffs and service quality provided by the Fair-Trading Commission (FTC).

45. External regulatory oversight rests with national authorities. The PDD manages development screening and permitting, with final permissions issued by the Planning and Development Board (PDB). For projects with potentially significant impacts, technical review is carried out by the multi-agency EIA Committee, which includes the Environmental Protection Department (EPD), the Coastal Zone Management Unit (CZMU), and the Ministry of Health and Wellbeing. However, the types of works anticipated under the Program, like NRW reduction measures, metering, and small sewer pilots, are small in scale and located within existing rights-of-way; therefore, they are unlikely to require a full EIA.

[2] The Barbados Water Authority (Amendment) Act, 2023 (amending Cap. 274A) provides for the temporary alteration of conditions attached to abstraction licenses during officially declared drought events, upon recommendation of the Water and Wells Management Committee.

[3] The WRM module under DLR 3.2 will be implemented through a Water Resources Information Management System (WRIMS), envisioned as an integrated system for managing and visualizing water resources information, linked to associated decision-support tools and other relevant systems managed by BWA for WRM. WRIMS will also enables data-sharing and integration with relevant national agencies involved in water resources management.

46. BWA has prior experience implementing projects financed by multilateral development banks (MDBs) and bilateral agencies, which required compliance with international environmental and social frameworks. These include the Inter-American Development Bank (IDB)–financed Water Sector Resilience Nexus project and mains rehabilitation programs supported by the Caribbean Development Bank (CDB). Through these operations, it is expected that BWA has gained exposure to MDB requirements in areas such as stakeholder consultation, grievance redress mechanisms, and the preparation and implementation of environmental management plans.

3. POTENTIAL ENVIRONMENTAL AND SOCIAL EFFECTS

47. The Barbados Water Security and Sector Performance Program is expected to generate overall positive outcomes in terms of improved service reliability, water conservation, and institutional strengthening. However, negative environmental and social impacts and risks must be identified, assessed, and managed appropriately. The following sections provide a summary of the identified impacts, risks, and benefits.

A. Exclusions

48. According to Paragraph 10 of the World Bank Policy on Program-for-Results Financing and Paragraph 14 of the World Bank Guidance: Program-for-Results Financing – Environmental and Social Systems Assessment, activities that are likely to have significant adverse impacts that are sensitive, diverse, or unprecedented on the environment and/or affected people are ineligible for PforR financing.

49. Such activities are excluded to ensure that the Program operates within a manageable environmental and social risk profile consistent with Core Principles 1–6 of the PforR Policy.

50. Based on this guidance and the scope of the Barbados Water Security and Sector Performance Program, the following activities are excluded from Program financing:

a. Works requiring land acquisition, physical displacement, or involuntary resettlement;
b. Works located within or adjacent to protected areas, marine reserves, coastal buffer zones, or other ecologically sensitive sites designated under national law or international conventions;
c. Activities causing significant conversion or degradation of critical natural habitats, such as mangroves, wetlands, or turtle nesting beaches;
d. Works situated in areas prone to severe natural hazards (coastal erosion, landslides, or flooding) unless robust engineering and climate resilience measures are fully integrated in design;
e. Works involving asbestos-containing materials, lead-based paints, or other hazardous or prohibited substances or technologies;
f. Activities that could adversely affect physical cultural heritage, including disturbance or destruction of antiquities, relics, or burial sites;
g. Activities with transboundary environmental implications (e.g., marine or groundwater pollution across national boundaries); and
h. Any intervention that, based on screening, would be classified as High risk under the World Bank Environmental and Social Framework (ESF) or equivalent national standards.

51. The Program must include a screening mechanism, to be institutionalized within the BWA WREMU, that confirms all proposed investments comply with these exclusions prior to financing. Any activity falling under these categories will be screened out during preparation and replaced by eligible interventions of comparable value and scope.

B. Potential Environmental and Social Benefits

52. Overall, the Program is designed to deliver broad environmental, social, economic, and institutional benefits, improving the quality of life for consumers, supporting sustainable development, and fostering greater equity and resilience within the community.

53. The Program delivers key environmental benefits by reducing water losses through leak detection and improved metering, which limits groundwater extraction and enhances water security. Efficiency measures lower energy consumption and the carbon footprint of water supply. Additionally, the sewer pilot reduces groundwater contamination risks by minimising reliance on on-site sanitation. Collectively, these initiatives support more sustainable water management and increase resilience to climate variability. Specific environmental benefits include:

a. *Water Resources Conservation:* The Program focusses on reducing NRW through leak detection, repair, pressure management, and loss control, along with improved metering. These measures collectively decrease physical water losses in the distribution system, which in turn limits the need for groundwater extraction. Sustained NRW control not only ensures a more reliable water supply for customers but also contributes significantly to long-term national water security.

b. *Improved Efficiency of Water Distribution Systems:* Measures such as more accurate metering and improved leak control leads to reduced physical water losses and decreases the need for excessive pumping. These actions not only optimize energy usage but also result in sustained long-term efficiency gains and help lower the carbon footprint of the water supply system.

c. *Reduced Risk of Groundwater Contamination:* The sewer pilot aims to reduce the risk of groundwater contamination by decreasing dependence on on-site sanitation systems and informal waste disposal. This initiative initially provides localized benefits by lowering the likelihood of untreated effluent seeping into soils and groundwater, with the potential for broader impact if successfully scaled up.

d. *Strengthened Water Security and Integrated Water Resources Management:* The Program strengthens water security and integrated water resources management by implementing targeted initiatives such as NRW reduction through leak detection and repair, pressure management, enhanced metering, a sewer pilot, and operational modernization. These measures collectively improve the reliability and sustainability of water supply, support better measurement and monitoring, bolster institutional capacity for evidence-based decisions, and increase resilience to droughts and climate variability, while also reducing the risk of groundwater contamination.

54. The potential social benefits include improved reliability and continuity of water services, better public health and hygiene, and temporary job creation during construction. The Program also aims to enhance staff skills and promote a safer, more inclusive work environment, particularly for women. Additional positive impacts include increased resilience to climate and disaster events, greater transparency and customer confidence in billing, and indirect economic gains in tourism due to improved water security. The Program delivers several significant social benefits including:

a. *Improved Reliability and Continuity of Water Services:* Upgrades to commercial services, such as the Customer Information System and Commercial Management System, support more consistent water supply and transparent billing practices. This reduces disputes and reinforces consumer trust in water charges.

b. *Enhanced Public Health and Hygiene:* By ensuring a reliable supply of potable water and better wastewater management through sewer pilots and pollution control, the Program helps to mitigate health risks and fosters safer, cleaner communities.

c. *Temporary Job Creation:* Construction and implementation phases provide short-term employment opportunities, contributing to local economic development and supporting livelihoods during project execution.

d. *Better Jobs through Capacity Building:* The Program offers technical and leadership training, adopts fair policies, and provides facilities like lactation rooms and separate bathrooms. It aims to boost staff skills, support career growth, and create a safe, female-friendly, and respectful workplace.

e. *Climate and Disaster Resilience:* Infrastructure modernization and improved water management practices increase the community's ability to withstand and recover from adverse events linked to climate change and natural disasters.

f. *Improved Customer Service and Transparency:* The digitalization of commercial services leads to more open and accurate billing systems, reducing the potential for disputes and enhancing customer confidence in the fairness of water charges.

g. *Gender Empowerment and Inclusion:* Through capacity building and the adoption of institutional policies, the Program actively promotes the involvement of women in technical and managerial roles within the Barbados Water Authority (BWA). This results in enhanced professional development, career progression for female staff, improved gender balance, and a more diverse and effective organisational culture.

h. *Tourism and Local Economic Benefits from Improved Water Security:* By reducing Non-Revenue Water (NRW), improving service reliability, and controlling pollution, the Program indirectly boosts tourism competitiveness. A reliable supply of safe water and improved wastewater management strengthen the tourism sector and generate long-term local economic benefits, including job creation and increased revenue.

55. Annex 2 contains detailed descriptions of the Program’s potential environmental and social benefits.

C. Potential Environmental and Social Risks, Impacts, and Mitigation Measures

56. Potential environmental risks and impacts associated with the Program are minor and the impacts are generally short-term, localized, and manageable with appropriate mitigation measures. Specific risks and impacts include:

a. *Vegetation Loss:* Vegetation and minor soil disturbance may occur during trenching and pipe-laying for the sewer pilot, potentially impacting local aesthetics and causing limited erosion. Mitigation measures include surveying and marking vegetation, minimising removal, restoring disturbed areas, and ensuring compensatory replanting. Soil disturbance from excavation is localised and low-risk, addressed by careful planning, prompt reinstatement, erosion control, and re-vegetation of affected areas.

b. *Minor Soil Disturbance and Erosion:* Minor soil disturbances as well as localised dust, noise, and vibration, may occur during small-scale excavation, leak repair, metering, and sewer pilot works. These impacts are generally limited due to the shallow depth and urban settings, and can be effectively managed by planning works during dry periods, promptly reinstating surfaces, implementing simple erosion controls, and using measures to minimise dust and noise. Regular inspections and use of well-maintained equipment further help to mitigate any adverse effects.

c. *Dust, Noise, and Vibration:* Activities such as leak repair, metering, and sewer pilot works can result in dust emissions, elevated noise, and ground vibrations, with recommended mitigation measures including dust suppression, noise control, and advance notifications to nearby residents. Minor water pollution risks may arise from improper spoil or wastewater disposal, but these impacts are typically short-term and localised; proper collection, disposal, and use of silt fences are advised. Additionally, small-scale generation of solid, construction, and hazardous waste should be managed responsibly to prevent environmental harm.

d. *Minor Water Pollution Risks:* Leak repairs, metering, and sewer pilot works can lead to temporary dust, noise, vibration, minor water pollution, and small-scale waste generation. These impacts are short-term and localised, with recommended mitigation including dust suppression, noise control, proper wastewater disposal, and waste segregation. Training for contractors and advance notifications for residents are advised to minimise environmental harm.

e. *Small-Scale Waste Management:* Ensuring proper waste management practices and risk mitigation measures for sewer pilot works and well monitoring installations. It emphasises the segregation and safe handling of construction and hazardous waste, proper disposal, recycling where possible, and contractor training. For monitoring equipment installation in wells, it details precautions to maintain well integrity and water quality, such as using dedicated observation wells, upgrading sanitary seals, disinfecting wells, and securing cables to prevent operational issues.
f. *Production Well Integrity and Water Quality Risks from Monitoring Installation:* Proper installation and maintenance of well monitoring equipment are crucial to prevent sanitary risks, operational disruptions, and water quality issues. Key practices include using dedicated observation wells, ensuring sanitary seals and waterproof connectors, disinfecting wells, and keeping thorough records. Additionally, works should be scheduled to minimize service interruptions. Temporary access disturbances, such as blocked sidewalks or driveways, may occur during activities like leak repair and metering, but these impacts are generally minor and short-term, especially in crowded urban environments.

57. The social risks associated with the Program are also limited in nature, generally short-term, and easily mitigated with proper planning, communication, and capacity building. Specific social risks and impacts include:

a. *Temporary Access Disturbances:* Minor and short-term access disturbances may occur during leak repairs, metering, and sewer pilot works, particularly in dense urban areas, due to trenching or excavation near property entrances. Measures such as advance notification, safe walkways, clear signage, and scheduling outside peak hours are recommended to reduce disruption to residents, businesses, and community facilities. Additionally, business livelihood impacts can be mitigated by briefing shop owners, providing temporary vending locations, and prioritising restoration of access points.
b. *Minor and Short-term Business Livelihood Disruption:* Program related Construction works, leak repair, metering, and the sewer pilot are expected to cause temporary business disruption, particularly for small shops and roadside vendors, as well as minor traffic and mobility impacts such as lane closures and congestion. Mitigation measures include engaging directly with affected businesses, providing clear signage and alternative vending spots, prioritising the restoration of access, and implementing approved traffic management plans with safe pedestrian routes.
c. *Minor and Short-term Traffic and Mobility Impacts:* Construction activities, leak repairs, metering, and the sewer pilot may cause temporary business, traffic, and mobility disruptions, which will be mitigated through direct engagement, signage, access restoration, and approved traffic management plans.
d. *Cultural Heritage Risks:* While the risk of disturbing cultural heritage is low since all works will take place in previously developed areas, strict chance-find procedures will be enforced if subsurface materials are encountered.
e. *Occupational Health and Safety Risks:* Work activities for metering and sewer pilot projects expose workers to occupational risks, such as traffic hazards, machinery accidents, and heat stress, necessitating strict OSH standards, PPE, and climate-adapted safety measures.
f. *Community Health and Safety Risks:* Hazards and mitigation measures associated with sewer pilot projects, such as risks from unsecured trenches, construction machinery, and community interaction, including the potential for SEA/SH. Mitigation includes inclusive community engagement, strategies for worksite safety, traffic management, dust and noise control, and waste handling.
g. *Digital Divide, i.e. the Exclusion of Vulnerable Users:* Efforts to modernise commercial services and water management include the adoption of digital platforms and operational committees, which raise concerns about accessibility for vulnerable users. Hybrid service delivery, inclusive design, targeted outreach, and fallback mechanisms are recommended to ensure equitable access, while transparency, early engagement, and support are key to addressing private users' concerns regarding data and monitoring procedures. Monitoring and feedback channels will help identify and resolve issues during implementation.

h. *Concerns from Private Users and Operators:* Private users may have concerns about new data and monitoring requirements as water management committees become operational, but clear communication, transparency, and support can mitigate potential conflicts. Additionally, the removal of informal water connections as part of NRW reduction measures may cause temporary disruption, with strategies in place to notify affected users and provide pathways for regularisation and service continuity.
i. *Disruption and Elimination of Informal or Unmarked Connections:* Measures are in place to address the impacts of removing illegal or informal water connections during NRW reduction and metering works, including pre-works surveys, communication, regularisation options, temporary service continuity, and grievance mechanisms.
j. *Service disruptions due to underground Utility Works:* There are risks of accidental damage to underground infrastructure during sewer and metering works, which can be mitigated through careful planning, consultation, advanced detection methods, and clear emergency protocols.
k. *Potential Opposition or Disruption Among BWA Staff:* Organizational reforms within BWA may face resistance, which is being managed through transparent communication, staff engagement, and capacity-building strategies.

58. Annex 3 contains detailed descriptions of the Program's potential environmental and social risks, impacts, and mitigation measures.

4. NATIONAL SYSTEMS ASSESSMENT

59. In line with the World Bank's PforR Policy, the assessment of program systems considers the institutional, organizational, and procedural arrangements in place for managing environmental and social effects. This includes the laws, regulations, policies, and implementing guidelines applicable to the Program, as well as the mandates of the institutions responsible for applying and enforcing them. The assessment also takes into account inter-agency coordination where responsibilities for oversight and compliance are shared across entities.

60. The analysis was undertaken to evaluate whether the existing national and BWA systems are adequate to manage the range of potential risks and impacts identified under the ESSA.

A. National Legal Framework

61. Barbados maintains a comprehensive body of legislation that governs environmental, social, labor, and health aspects relevant to water and sanitation projects. This framework combines overarching development control instruments—such as the Planning and Development Act (2019, amended 2020) and Environmental Impact Assessment Regulations (2021)—with sectoral laws addressing environmental protection, coastal and natural resource management, cultural heritage, public health, labor rights, and workplace safety. Together, these instruments establish the legal authority for environmental and social screening, permitting, compliance, and enforcement. Annex 4 summarizes the main elements of the national legal framework and their applicability, highlighting the extent to which they provide safeguards for Program activities.

62. Barbados's key environmental laws and regulations relevant to the Program and summarises their applicability. The Planning and Development Act 2019 and the Environmental Impact Assessment (EIA) Regulations 2021 set out procedures for development control, including requirements for environmental review, particularly for projects with significant potential impacts. For the Program's activities—such as non-revenue water (NRW) reduction, metering improvements, and small sewer pilots—full EIAs are typically not needed due to the limited scale and use of existing rights-of-way, but further assessment may be required if works are close to sensitive areas like heritage or conservation sites.

63. The Environment Protection Act 1997 provides the framework for environmental management, empowering authorities to regulate pollution, emissions, and waste. Program activities must comply with these rules, especially regarding the management of construction waste and hazardous materials (such as oils and asbestos), to prevent environmental harm.

64. With regards to care needed for Program related works, the Electric Light & Power Act, Cap. 278, and the National Petroleum Corporation Act, Cap. 280, mandate that the Program must coordinate with the respective utility licensees, obtain their consent, and comply with safety and technical conditions before executing works near electricity or gas lines. The Road Traffic Act, Cap. 295, requires securing permits for any works affecting public roads, implementing approved traffic management plans, and ensuring prompt restoration of roadways and sidewalks. Additionally, the Preservation of Antiquities and Relics Act, 2011 governs the identification, protection, and management of antiquities, cultural heritage, outline the responsibilities of relevant authorities and the legal obligations for safeguarding these assets. Collectively, these laws ensure that the Program operates safely, legally, and with minimal disruption to public services and safety.

65. The legal and regulatory framework for water and sanitation management in Barbados is established through several acts including the Barbados Water Authority Act, 1980 empowers the BWA to oversee water resources, implement key activities such as non-revenue water (NRW) reduction and metering, and manage infrastructure, ensuring the Program operates with clear institutional authority. The Utilities Regulation Act, 2002, and its 2010 Amendment, place the BWA under the Fair Trading Commission's regulatory oversight, ensuring transparency, consumer protection, and proper tariff setting, which supports the Program's financial and operational sustainability. Finally, the Barbados Water Authority (Water Services) Regulations, 1982, define the obligations of both the BWA and consumers regarding water distribution, supply, metering, and billing, providing practical guidelines for the Program's day-to-day implementation and consumer engagement. The Water Reuse Act, 2023 (Act 2023-10) Creates a permit system to produce, supply, or sell reclaimed (non-potable) water. Establishes inspection, enforcement, suspension/revocation, offences, and administrative penalties. Establishes a Water Reuse Committee to advise on permits and compliance. Defines allowable uses of reclaimed water (e.g., irrigation, equipment cooling, groundwater recharge). The Act aims to promote sustainable water management by regulating the safe and beneficial use of reclaimed wastewater.

66. The workplace health and safety legislation relevant to the Program primarily consists of the Safety and Health at Work Act, 2005 (as amended in 2022) and its subsidiary regulations. This framework obliges employers, including BWA and its contractors, to ensure safe working conditions through risk assessments, worker training, provision of personal protective equipment (PPE), and adequate sanitary facilities. The 2022 Amendment broadened employer and worker responsibilities and strengthened inspectorate powers, aligning local standards with international conventions. Specific regulations address risks such as noise exposure, hazardous materials, heavy equipment operation, and heat stress, requiring targeted preventive measures, monitoring, and compliance with official directives. Additionally, the Employment Rights Act, 2012-19 guarantees fundamental labour rights for all staff and contractors, including fair terms of employment and protections against unfair dismissal, supporting the overall objective of safeguarding health, safety, and welfare in the workplace.

67. The Land Acquisition legislation relevant to the Program sets out the legal framework for the compulsory purchase of land required for public works, such as those undertaken by the Barbados Water Authority (BWA) and its contractors. It outlines the procedures for identifying affected properties, notifying landowners, and determining fair compensation for acquired land. The legislation ensures that acquisitions are conducted transparently and in accordance with due process, safeguarding the rights of property owners while enabling essential infrastructure projects to proceed efficiently. This framework is crucial for minimising disputes and facilitating timely delivery of Program works.

68. In addition to the legal framework there are two key national initiatives in Barbados related to the Program, the Integrated Solid Waste Management Program (ISWMP), and the Comprehensive Disaster Management (CDM) Policy. ISWMP, supported by international partners, aims to modernise waste management practices, covering landfill operations, recycling, hazardous waste, and institutional capacity, though it is not legally binding. The CDM Policy provides strategic guidance for integrating disaster risk reduction and climate resilience in the water sector, encouraging the incorporation of emergency preparedness and resilient design. Both initiatives offer important frameworks for sustainable development, despite not carrying legal obligations.

B. International Legal Framework

69. Barbados is party to several international conventions and treaties that are relevant to the Program. The majority of these aim to safeguard environmental and occupational health and safety standards across borders. A comprehensive list of and analysis of how they relate to the Program is found in Annex 5. They serve as authoritative guidance for marine protection, hazardous waste management, and occupational health and safety, each providing specific obligations and recommended practices for member states. Collectively, these agreements establish minimum standards and procedures designed to ensure responsible stewardship of natural resources and the safety of workers.

70. For instance, the Cartagena Convention offers a comprehensive structure for the protection and sustainable use of marine and coastal environments in the Wider Caribbean Region. Its protocols address issues such as land-based sources of pollution, conservation of specially protected areas and wildlife, and oil spill response. Similarly, the Basel Convention regulates the international movement and disposal of hazardous wastes, requiring strict controls and prior informed consent to prevent environmental harm. These conventions are complemented by instruments like the ILO C155, which sets global benchmarks for workplace health and safety, further reinforcing the commitment to sustainable development and human well-being.

71. In practice, these international agreements not only set overarching legal requirements but also promote cooperation and information sharing among nations. They encourage the adoption of best practices in areas such as waste handling, pollution prevention, and occupational risk management. For programs operating within Barbados, adherence to these frameworks ensures alignment with global standards, reduces the risk of environmental degradation, and supports the creation of safer workplaces. This legal context ultimately underpins the responsible execution of development activities, contributing to the long-term health of both people and the environment.

C. National System Capacity to Respond to Identified Environmental and Social Risks

72. As described above and detailed in Annexes 4 and 5, Barbados' legal framework provides comprehensive authority and clear institutional responsibilities for addressing environmental and social risks and impacts relevant to the Program. However, in preparing the ESSA some implementation gaps were identified that will need to be addressed during program implementation. The major recommendations are included in the PAP. Annex 6 contains the detailed analysis done for making the overall recommendations and preparing the environmental and social actions in the PAP.

73. The Barbados Water Authority Act and Water Services Regulations lay out BWA's responsibilities for asset management, lawful connections, service continuity, and the procedures for planned service interruptions, ensuring a baseline of operational stability and public notification. This helps mitigate environmental and social risks related to service disruptions, such as lack of access to essential water services.

74. Sewerage Regulations require formal submissions and approvals for sewer works, enforcing system compliance and promoting inter-agency coordination. This is crucial for minimizing public health risks and environmental hazards, particularly in densely populated areas. The regulations also support the prevention of unsanitary conditions in the event of infrastructure strikes or failures, as further reinforced by the Health Services Act and Nuisances/Building Regulations, which empower authorities to act when public health is at risk due to utility-related incidents.

75. Permit requirements under the Road Traffic Act and Highways Act, along with the necessity for Traffic Management Plans, ensure that road works are conducted safely and with minimal disruption to the public. These measures address social risks such as restricted mobility, increased accident risks, and the economic impact of prolonged road closures. Similarly, crossing or working near electric and gas lines requires adherence to the respective licensee's safety standards and explicit consent, protecting both workers and the general public from potential hazards.

76. The Planning and Development Regulations provide for the imposition of protective conditions, such as method statements and work sequencing, which can extend to cross-utility coordination and approvals. This fosters a culture of safety and accountability, reducing the risk of accidental strikes and service interruptions that could negatively affect

communities and businesses. However, the effectiveness of these mechanisms depends on robust enforcement and active collaboration among utility providers.

77. Despite the strengths of the national system, there are notable gaps, particularly related to the absence of reviewed Standard Operating Procedures (SOPs) for BWA. Formalizing and approving SOPs should include construction management; maintenance operations; site and dig safety; waste management including segregation, labelling, temporary staging, collection, and reuse-recycling; schedules; record-keeping; are either unclear or undocumented for these types of waste. Such actions would improve transparency and community outreach; consistency in managing third-party consents; wayleaves; pre-dig coordination.

78. Additionally gaps in practical implementation were noted within the BWA concerning small-scale, dispersed construction wastes. Specific operational procedures

79. Contractor orientation is another area of concern, as there is no evidence of training or requirements for housekeeping, litter control, or secure storage measures to prevent windblown debris and scattering at construction sites. This oversight increases the risk of environmental harm, particularly from small quantities of waste generated during rolling works, which may not be subject to the same scrutiny as larger projects.

80. Finally, the existing grievance and communication channels are primarily designed for billing and service issues, with no mechanism in place for advance public notices or for receiving and resolving complaints specifically related to construction-phase activities. Within BWA, efforts are needed around the formalization and standardization of utility coordination and strike prevention protocols.

81. To address these gaps, the PAP recommends updating BWA's SOPs to align fully with national regulations and to include a formalised process for recording utility consents and wayleaves, conducting coordinated pre-dig activities, and enforcing protection standards at utility crossings. The SOPs should also establish clear protocols for utility-strike response, including mark-out standards and safe-dig specifications. Enhanced competency-based training for supervisors and plant operators, along with the introduction of field inspection checklists and key performance indicators, will further strengthen compliance and reduce social risks associated with utility works.

82. In summary, the Barbados national system provides a solid foundation for managing the social risks linked to utility works, with multiple layers of regulation and inter-agency coordination. However, improvements in SOP transparency, process standardisation, and workforce training are necessary to close existing gaps and ensure that social and environmental risks are managed proactively and effectively.

83. The Program, including the PAP, aims to enhance both the national and BWA systems' capacity to manage environmental and social risks effectively.

5. CONCLUSIONS AND RECOMMENDATIONS

A. Alignment of National Environmental and Social Systems

84. The World Bank's PforR policy outlines six guiding principles for managing the environmental and social aspects of the programs it finances. The assessment of the environmental and social systems applicable to the Program indicates that Barbados's legal and regulatory framework is generally aligned with the core principles and planning elements of the PforR Directive. This framework adequately addresses environmental, social, and occupational health and safety considerations. However, certain gaps have been identified, along with opportunities for institutional strengthening.

85. Key recommendations to BWA based on the World Bank's six PforR environmental and social Core Principles are detailed below.

86. Core Principle #1: Promoting Environmental and Social Sustainability - Program environmental and social management systems are designed to promote sustainability within the Program's design, avoid or mitigate adverse impacts, and support informed decision-making regarding environmental and social effects. Recommendations include:

a. *Early Screening of Potential Impacts*. Barbados' Planning and Development Act and EIA Regulations require systematic project screening to determine if an IEE or full EIA is needed, with criteria including location and potential impacts. Small-scale utility works, such as those in the Program, are usually exempt or require only IEEs unless near sensitive sites. The Barbados Water Authority's WREMU provides technical guidance, and SOPs exist for managing construction impacts; however, there is no consistent tool to screen all environmental and social risks. It is recommended to adopt a checklist for all activities and exclude projects with significant adverse impacts not aligned with World Bank PforR policies.

b. *Consideration of Strategic, Technical, and Site Alternatives.* The Planning and Development Act (2019) and EIA Regulations (2021) require EIAs to include analysis of project alternatives, covering technical approaches, site locations, and a "no development" scenario. Most Program activities are unlikely to trigger a full EIA and may only require an IEE, which does not consistently require detailed consideration of alternatives. Recommendations involve reviewing or developing BWA internal guidelines for considering alternatives for all Program activities and institutionalizing documentation of the "no action" scenario. Strengthening the link between sectoral planning tools and BWA decision-making is also advised.

c. *Assessment of Induced, Cumulative, and Transboundary Impacts*. EIA Regulations (2021) require EIAs to evaluate significant effects, including interactions between impacts, providing a basis for considering cumulative impacts. However, comprehensive cumulative impact assessment and strategic environmental assessment are not explicitly mandated. For small-scale works, induced, cumulative, or transboundary impacts are unlikely, though localized overlaps could occur. Recommendations include integrating questions on overlaps in timing and location into BWA's screening tool, requiring contractors to prepare traffic and waste management plans, and ensuring marine safeguards for sewer pilots near coastal areas.

d. *Identification of Mitigation Measures*. Barbados' legal framework mandates identification and implementation of mitigation measures through EIA Regulations and sectoral laws such as the Environment Protection Act, Coastal Zone Management Act, Safety and Health at Work Act, Health Services Act, and Road Traffic Act. These frameworks require pollution control, erosion management, occupational health and safety, and traffic safety measures. Mitigation is generally required when projects undergo an IEE or EIA, with systematic mitigation for small-scale interventions not automatically activated unless imposed by authorities. Recommendations include developing standardized Environmental Codes of Practice (ECoPs) and SOPs, integrating them into contracts, enhancing monitoring capacity, and providing training for staff and contractors.

e. *Articulation of Institutional Responsibilities and Resources*. The Planning and Development Department (PDD), Planning and Development Board (PDB), and the EIA Committee form the core regulatory mechanism for integrating environmental and social considerations into project approvals. BWA has an Environmental Unit, but lacks a documented, Program-wide definition of roles and resources to implement plans. There is no evidence of standard ECoPs or integrated social SOPs. Recommendations focus on developing clear Program-level procedures, providing capacity building, and designating environmental and social focal points within BWA.

f. *Responsiveness and Accountability through stakeholder consultation and grievance Redress mechanisms*. Legal provisions require public notice, opportunities for written submissions, and public hearings for projects requiring IEE or EIA. At the Program level, BWA engages with customers but lacks mechanisms specifically addressing environmental and social issues arising from works. Grievance handling exists through consumer complaint processes and statutory pathways but is not tailored to construction-phase risks. Recommendations include standardizing consultation and disclosure for all works, establishing a Program-level grievance redress mechanism (GRM), embedding consultation and GRM requirements in contracts, setting service standards and escalation protocols, and providing resources and training for implementation.

87. Core Principle #2: Protection of Natural Habitats and Cultural Resources - Program environmental and social management systems aim to avoid, minimize, or mitigate adverse impacts on natural habitats and physical cultural resources. Activities involving significant conversion or degradation of critical natural habitats or cultural heritage are not eligible for PforR financing. Recommendations include:

g. *Identification and Mitigation of Impacts on Biodiversity and Cultural Resources*. Barbados' legal framework includes instruments for screening and mitigating potential impacts on natural habitats and cultural resources, with specific controls and conditions imposed by regulatory agencies. Most Program activities are unlikely to trigger significant impacts, but localized risks may occur. Screening is reactive, with no standardized procedure within BWA. Recommendations include developing screening checklists and ECoPs, strengthening collaboration with technical authorities, using biodiversity mapping tools, training staff and contractors, establishing procedures for chance finds, and excluding high-risk activities from the Program.

h. *Protection, Conservation, and Rehabilitation of Natural Habitats*. Regulatory instruments such as the Coastal Zone Management Act, Physical Development Plan, Trees (Preservation) Act, Soil Conservation Act, and National Conservation Commission Act provide statutory mechanisms for habitat protection. Program activities are expected to occur mainly within urban rights-of-way, but risks may arise near sensitive areas. Proactive conservation measures are not systematically embedded, and monitoring is limited. Recommendations include introducing standardized screening tools, excluding high-risk activities, enhancing monitoring, and providing capacity building.

i. *Avoidance of Significant Conversion or Degradation of Critical Natural Habitats*. Program works are expected in urban areas, with direct interaction with critical habitats unlikely. However, interventions extending into sensitive areas are subject to biodiversity studies and specialized surveys. No mechanism exists within BWA for verifying potential overlaps before works are designed. Recommendations focus on excluding high-risk activities from financing.

j. *Protection of Physical Cultural Property.* National legislation prohibits unauthorized excavation or removal of antiquities and integrates heritage into development control. The likelihood of direct impacts on cultural resources is low, but excavation could encounter relics. Current procedures rely on permitting triggers, with no systematic screening tool for managing such risks. Recommendations include developing screening tools, integrating chance-find procedures into SOPs and contracts, providing training, and incorporating standard heritage protection clauses.

88. Core Principle #3: Safety of Public and Workers - Program environmental and social management systems are designed to protect public and worker safety during construction, operation, and maintenance, and from exposure to hazardous materials or natural hazards. Recommendations include:

a. *Community, Individual, and Worker Health, Safety, and Security*. Legislation covers occupational and community safety, with mandatory risk assessments, training, and protective equipment. No published standard OSH/EHS procedures under BWA were identified, nor formalized contractor guidance. Recommendations include developing and enforcing standard Environmental, Health, and Safety Codes of Practice, incorporating mandatory clauses into contracts, training staff and contractors, strengthening supervision, and establishing incident response protocols.

b. *Child and Forced Labor*. Legislation prohibits child and forced labor, with minimum working age set at 16 years. World Bank standards require no employment of minors under 18 in hazardous activities. No explicit contractual clauses were identified. Recommendations include including standard labor clauses, verifying the age of workers, collaborating with the Labor Department for inspections, and providing orientation and training.

c. *Management of Hazardous Materials*. Regulation is provided by the Environment Protection Act and other sectoral laws, but systems are designed for large-scale waste streams and do not address small-scale hazardous waste typical of Program activities. No Program-specific procedures or training protocols were identified. Recommendations include developing standardized protocols, strengthening internal guidance, building contractor capacity, formalizing incident response, and clarifying safe disposal pathways.

d. *Integrated Pest Management*. Pesticide control is regulated for agricultural use, but there are no measures promoting integrated pest management for infrastructure works. No guidance or training exists within BWA. Recommendations include prioritizing non-chemical pest management measures and developing contractor guidance notes.

e. *Worker Training for Hazardous Chemicals*. General obligations for risk assessment and training exist, but specific protocols for hazardous materials are lacking. Recommendations include developing tailored training modules, integrating international good practice references, and requiring contractors to document worker orientation.

f. *Risk Management in Areas Prone to Natural Hazards*. The Emergency Management Act provides the national framework for preparedness, with sectoral laws addressing specific risks. Program activities are not expected to be highly vulnerable, but construction phase risks exist. No specific procedures for integrating hazard preparedness were identified. Recommendations include introducing hazard-screening steps, incorporating resilience measures, and developing contractor guidance notes.

89. Core Principle #4: Managing Land Acquisition and Access to Natural Resources - Program systems aim to avoid or minimize displacement and assist affected people in restoring or improving livelihoods and living standards. Recommendation include:

a. *Land Acquisition and Impacts*. Activities requiring involuntary land acquisition or physical displacement are ineligible for financing. Program investments are expected within existing public rights-of-way, with only temporary access restrictions anticipated. No systematic procedures exist for screening or mitigating short-term disruptions. Recommendations include clarifying exclusion of activities requiring land acquisition, screening for access restrictions, minimizing disruptions through contractual requirements, and ensuring accessible grievance mechanisms.

90. Core Principle #5: Attention to Vulnerable Groups and Equitable Access - Program systems must consider the cultural appropriateness of benefits, equitable access, and the needs or concerns of vulnerable groups. No Indigenous Peoples are present in Program areas. Exclusion risks for other vulnerable groups are assessed under Core Principle 1. No further action is required.

91. Core Principle #6: Avoiding Social Conflict - Program systems must avoid exacerbating social conflict, particularly in fragile or post-conflict areas. Recommendations include:

a. *Management of Conflict Risks*. The Program could potentially exacerbate labor conflicts due to institutional changes. Barbados' labor legislation recognizes the right to unionize, and BWA employees are represented by the Barbados Workers' Union. Formal communication and engagement strategies incorporating union participation have not been developed. Recommendations include establishing structured participation mechanisms and procedures for engaging workers' representatives as part of the broader communication and engagement strategy during the modernization process.

92. See Annex 7 for the full analysis of the core principles by assessing the extent of the alignment, identifying gaps and providing related recommendations.

B. Elements to Integrate into the Program Action Plan

93. To assist in operationalizing these broader recommendations, the following four environmental and social items are included in the Program Action PAP:

a. Boosting BWA's environmental and social capacity;
b. Developing standard environmental and social protocols and standardizing SOPs;
c. Launching a stakeholder engagement plan the Program; and

d. Standardizing community engagement procedures to include grievance monitoring and redress for construction activities.

94. The four environmental and social actions contained in the PAP, as detailed in Table 4, below, are based on the recommendations made in the ESSA. In carrying out these actions, BWA is expected to adhere to those recommendations as outlined above and included in Annexes 3, 6, and 7.

Table 4 – Environmental and Social Actions for the Project Action Plan

	PAP Item	Source	Responsibility	Timing	Completion Measurement
1	Boost BWA's capacity by hiring or appointing E&S risk management and gender experts. And training relevant BWA staff in environmental and social risk mitigation, community health and safety, Occupational Health and Safety, and SEA/SH protocols.	ESSA	BWA	December 31, 2026	Specialists are engaged or appointed for the Program with the required skills per agreed Terms of Reference. Initial BWA training has been conducted, with attendance documented and materials archived. Additional training will be held throughout the program.

	PAP Item	Source	Responsibility	Timing	Completion Measurement
2	Develop standardized E&S Codes of Practice, Standard Operating Procedures for small-scale water and sanitation works, and a screening. with the Bank to identify E&S risks and requirements for inclusion in bidding documents.	ESSA	BWA	Prior to the first tender launch or within 6 months after Program effectiveness, whichever comes first.	Environmental Codes of Practice, Standards Operating Procedures and screening checklist developed and adopted by BWA. BWA's environmental and social requirements are incorporated into all civil works bidding and contracting documents.
3	Launch a stakeholder engagement plan for the overall Program, including digitalization, septage and wastewater reuse uptake, and groundwater management.	ESSA	BWA	December 31, 2026	Provision of completed Stakeholder Engagement Plan are developed and satisfactory to the Bank.
4	Standardize consultation and disclosure procedures for all water and sanitation small works that includes a Program-level incident reporting and grievance mechanism for stakeholders to report issues related to implementation.	ESSA	BWA	December 31, 2026	Procedures for consultations and disclosure of water and sanitation small works including incident reporting and grievance mechanism adopted and implemented by BWA.

ANNEX 1 – SUMMARY OF STAKEHOLDER CONSULTATIONS

95. The Barbados Water Authority (BWA) organized stakeholder consultations on the draft Environmental and Social Systems Assessment (ESSA) for the Barbados Water Security and Sector Performance Program (P508674) (the Program) on December 9, 2025. A second stakeholder consultation both on the Program and the draft ESSA was held on January 20, 2026. Internal and external BWA Stakeholders were included in both consultations including all BWA Departments, Unions representing the water sector, the Prime Minister's Office, Ministry of Finance, Ministry of Health, and Ministry of Environment. A summary of those discussions follows.

96. Comments and clarifications received during these consultations enabled the ESSA Team to better understand how the environmental and social compliance works on the ground, identify missing elements, and clarify the recommendations in order to develop a robust and realistic Project Action Plan (PAP).

97. **Project Design and Disbursement Linked Indicators (DLIs):** The World Bank team presented the overall technical design of the water sector Program, focusing on the structure, objectives, and the five main DLIs, while BWA participants raised questions about targets, funding, and implementation details.

a. **Program Structure and Objectives:** the Program's high-level objectives, including water security, pollution management, and improved performance of BWA, with a $50 million loan based on BWA's strategic plan were outlined. The Program is part of a regional initiative, with Barbados as the front runner, and includes plans for a regional center of water excellence.

b. **DLIs Overview and Flexibility:** It was explained that the Program is structured around five DLIs, each with sub-indicators and scalable targets, allowing for partial disbursement as milestones are achieved. The DLIs cover areas such as reducing water losses, commercial improvement, institutional strengthening, wastewater management, and water security.

c. **Funding Allocation and Value for Money:** The allocation of funds to each DLI is based on the relevance and value for money rather than direct costs, with flexibility for BWA to reallocate funds as needed, provided overall targets are met. The Bank presented a breakdown of the US$50 million allocation, clarifying that these figures are not fixed and can be adjusted during implementation.

d. **Clarification on Metering and NRW Targets:** BWA participants questioned the feasibility of installing 80,000 smart meters and achieving a 25% non-revenue water (NRW) target, citing funding constraints and the need for realistic targets. After discussion, the Bank agreed to consider revising the NRW target to 30% and to clarify whether the target should include both physical and commercial losses.

e. **Implementation Arrangements:** The Program's implementation structure was outlined, including oversight by the Prime Minister's Office, the BWA board, and a transformation committee, with BWA responsible for hiring an independent verification agent and establishing a Project Execution Unit (PEU). BWA requested updates to reflect their internal nomenclature and committee structure.

98. **Environmental and Social Systems Assessment (ESSA) and Program Action Plan:** The Bank team led a review of the ESSA, highlighting the assessment of Barbados's environmental and social systems, identified gaps, and the proposed Program action plan, with BWA participants providing feedback and requesting clarifications on timelines and responsibilities.

a. **ESSA Objectives and Exclusions:** It was explained that the ESSA evaluates the adequacy of Barbados's legal, regulatory, and institutional frameworks for managing environmental and social risks under the Program, with exclusions for activities requiring land acquisition, resettlement, or work in protected areas.

b. **Identified Risks and Benefits:** The team outlined expected social and environmental benefits, such as improved service reliability and water conservation, as well as risks including temporary access disruptions, occupational health and safety, digital exclusion, and potential concerns from private well users and staff.

c. **Key Gaps and Recommendations:** Gaps identified include limited environmental and social staffing, fragmented monitoring systems, absence of standardized codes and practices, inconsistent consultation procedures, and lack of a documented grievance mechanism. Recommendations include hiring or assigning staff for environmental and social risk management, developing standardized procedures, and improving stakeholder engagement.

d. **Program Action Plan (PAP) and Next Steps:** The PAP includes four main actions: strengthening environmental and social capacity, developing standard operating procedures, launching a stakeholder engagement plan, and standardizing consultation and disclosure procedures. BWA requested draft terms of reference for new roles and clarification on the integration of these actions into their processes.

e. **Timelines and Document Review:** The World Bank team emphasized the need for BWA to review and provide feedback on the ESSA and Program action plan by the end of the week, with flexibility on timelines for implementation but a firm deadline for agreeing on the actions themselves. BWA highlighted the importance of consulting with the union and internal stakeholders before finalizing commitments.

99. **Stakeholder Engagement, Union Involvement, and Document Sharing:** BWA participants, including union representatives, raised concerns about receiving Program documents for the first time, requested additional time for review, and emphasized the need for further consultation and clarification on key indicators, gender targets, and implementation details.

a. **Document Access and Review Process:** BWA and union representatives noted that some documents, such as the Program document and NRW reduction plan, were being seen for the first time, and requested time to review and discuss them internally before providing feedback or agreement.

b. **Union Consultation and Clarification Requests:** Union representatives asked for clarification on gender balance targets, the scope of accreditation indicators, and the timeline for achieving these targets, with the Bank confirming that the gender accreditation target is 40% women by the end of the five-year Program and is scalable.

c. **Sharing of Supporting Documents:** The World Bank team committed to sharing slides, the ESSA, and other relevant documents with BWA and union representatives, and to providing examples and draft terms of reference for new roles as requested.

100. **Coordination with External Agencies and Next Steps:** The Bank and BWA agreed on coordination with external agencies such as the Ministry of Health, FTC, and PIU, and outlined next steps for finalizing Program documents, setting appraisal dates, and preparing for upcoming meetings.

a. **Coordination with Ministry of Health and FTC:** BWA confirmed participation in upcoming meetings with the Ministry of Health regarding septage management regulations and noted FTC's involvement in the Program, with the Bank welcoming their input.

b. **Finalization of Program Documents:** The Bank emphasized the need to finalize the DLIs and Program action plan by Friday to meet internal World Bank deadlines, with flexibility for further adjustments during appraisal and negotiation.

ANNEX 2 – POTENTIAL ENVIRONMENTAL AND SOCIAL BENEFITS

Potential Environmental Benefit	DLIs and Activities Likely to Enable It	Nature of Benefit (scope, duration, likelihood)
Water Resources Conservation	• DLIs 1, 3, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering.	NRW reduction measures reduce physical losses in the distribution network, thereby decreasing the need for groundwater abstraction. The benefits to both consistent water supply to customers and long-term national water security are cumulative, if NRW control measures are sustained over time.
Improved Efficiency of Water Distribution Systems	• DLIs 1, 3, and 5. • NRW reduction. • Metering.	More accurate metering and effective leak control reduce physical losses and the need for excessive pumping, leading to optimized energy use. These improvements generate long-term efficiency gains and contribute to reducing the overall carbon footprint of the water supply system.
Reduced Risk of Groundwater Contamination	• DLI 4. • Sewer pilot.	By reducing reliance on on-site sanitation systems and informal disposal practices, the pilot lowers the probability of untreated effluent infiltrating soils and groundwater. The benefits are initially localized, but may be scaled up if the approach is replicated.
Strengthened Water Security and Integrated Water Resources Management (WRM)	• DLIs 1, 2, 3, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering. • Sewer pilot. • Operational modernization and digitalization of commercial services.	Improved reliability and sustainability of water supply through better measurement, monitoring, and regulation of water resources. Enhanced data and institutional capacity enable evidence-based decision-making, reduced abstraction pressures, and improved resilience to droughts and climate variability.

Potential Social Benefit	DLIs and Activities Likely to Enable It	Nature of Benefit
Improved Service Reliability and Continuity	• DLIs 1, 2, 3, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering.	Improved reliability and continuity of water services, including a reduction in leaks, fewer unplanned service interruptions, and more consistent water pressure across the distribution network.
Enhanced Public Health and Hygiene	• DLI 4. • Sewer pilot.	Reduction of sewage overflows, safer disposal of fecal sludge, and improved sanitation access.
Job Creation (short term)	• DLIs 1, 3, and 4. • NRW reduction (leak detection and repair; pressure management; and loss control measures). • Metering. • Sewer pilot.	Potential generation of temporary employment opportunities during construction activities. Given that most works are carried out by BWA staff, the scale of this benefit may be limited, and the creation of permanent jobs is not anticipated.
Better Jobs through Capacity Building	• DLIs 1, 2, 3, 4. and 5 • Technical and leadership training programs for staff. • Equitable policies and improved work environment.	Strengthened staff capacities and technical skills, enabling personnel to perform higher-level functions, assume greater responsibilities, and access better professional opportunities. Promotion of a female friendly, safe work environment (lactation rooms, separate bathrooms), staff trained on respectful, harassment-free behavior.
Climate and Disaster Resilience	• DLIs 1, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering. • Sewer pilot.	Strengthened resilience of water and sanitation services to climate variability and extreme events. Enhanced system efficiency and reliability. Populations are less exposed to service disruptions during extreme climate- and disaster-related events.
Improved Customer Service and Transparency	• DLIs 1, 2, 3, and 5. • Metering and data flow to billing systems. • Operational modernization and digitalization of commercial services (Customer Information System / Commercial Management System),	More transparent billing, reduced disputes, and stronger consumer confidence in water charges.

Potential Social Benefit	DLIs and Activities Likely to Enable It	Nature of Benefit
Gender Empowerment and Inclusion	• DLI 3. • Promotion of gender inclusion in technical and managerial positions within BWA through capacity building and institutional policies.	Enhanced professional development and career advancement opportunities for women within BWA; greater gender balance in technical and managerial positions; improved organizational culture, diversity, and institutional performance.
Tourism and Local Economic Benefits from Improved Water Security	• DLIs 1, 4, and 5. • NRW reduction; sewer pilot (pollution control); service reliability.	Indirect, longer-term gains in tourism competitiveness and related jobs from reliable potable water and improved wastewater management.

ANNEX 3 – POTENTIAL ENVIRONMENTAL AND SOCIAL RISKS, IMPACTS, AND MITIGATION MEASURES

Potential Environmental Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Vegetation Loss (minor)	• DLIs 1, 4, and 5. • Sewer pilot.	Trenching and pipe-laying may require trimming or removal of roadside vegetation, ornamental plants, or small trees located along rights-of-way. This could affect local aesthetics, shade provision, and in some cases contribute to minor soil exposure or erosion.	• Conduct pre-construction surveys to identify and mark vegetation along rights-of-way. • Adjust trench alignment or excavation methods to avoid vegetation removal where feasible. • Limit trimming to the minimum necessary and avoid full removal unless absolutely required. • Restore disturbed areas after works, including replanting ornamental species or replacing removed trees. • Preserve and reuse topsoil during excavation to promote regrowth and reduce erosion. • Obtain permits and ensure compensatory replanting at a minimum ratio of two new trees for each protected or mature tree removed, prioritizing native or ecologically appropriate species, and maintain replacements until an 80–90 percent survival rate is achieved within 24 months. • Inform nearby residents and businesses about vegetation impacts and reinstatement measures.
Minor Soil Disturbance and Erosion	• DLIs 1, 4, and 5. • Sewer pilot.	Small-scale excavation for short pipe sections may cause limited soil disturbance or minor displacement of surface materials. Given the shallow depth and urban or previously modified settings, the potential for erosion or sediment runoff is low and localized, mainly during rainfall if trenches remain open. Proper sequencing and prompt reinstatement are sufficient to control these effects.	• Plan excavation during dry weather where possible. • Keep excavated soil stockpiled neatly away from drains or water channels. • Backfill and compact trenches promptly after pipe installation. • Use simple erosion-control measures (e.g., temporary barriers or silt fences) in sensitive areas. • Re-vegetate or restore disturbed verges and surfaces soon after completion. • Inspect reinstated areas after rainfall to address any localized erosion or settlement.

Potential Environmental Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Dust, Noise, and Vibration (localized)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Trenching, pipe laying, and movement of machinery and materials can generate dust emissions, elevated noise levels, and ground vibrations.	• Lightly spray exposed soil or excavation areas during dry or windy conditions to reduce dust. • Cover or dampen stockpiles and truck loads of excavated material during transport. • Limit work to daytime hours in residential or sensitive areas to minimize disturbance. • Use well-maintained equipment fitted with silencers or mufflers to reduce noise and vibration. • Avoid prolonged idling of machinery and sequence works to shorten exposure periods. • Notify nearby residents or institutions in advance of any short periods of elevated noise or vibration.
Minor Water Pollution Risks (sediment, debris, wastewater from pipe flushing)	• DLIs 1, 4, and 5. • Sewer pilot.	Improper disposal of spoil or flushing water may temporarily affect roadside drains. Impacts are short-term, localized, and unlikely to affect broader ecosystems.	• Collect and properly dispose of wastewater generated from pipe flushing in designated drains or disposal points approved by authorities. • Install silt fences, sediment traps, or barriers near drainage inlets to prevent runoff of spoil or debris. • Avoid direct discharge of flushing water, sediment, or waste into storm drains, waterways, or coastal areas. • Schedule flushing and dewatering activities during low-flow or dry weather periods to minimize downstream impacts. • Train contractors and crews on proper wastewater handling, disposal practices, and emergency spill response.
Small-Scale Solid, Construction, and Hazardous Waste Generation (old pipes, excavation spoil, packaging, broken meters, lubricants, asbestos fragments)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Works will generate limited amounts of construction and solid waste, including spoil, old pipes, packaging, and broken meters, as well as small quantities of hazardous materials (oils, lubricants, possible asbestos).	• Segregate construction, solid, and hazardous waste at the worksite. Collect and transport waste to approved SSA or municipal disposal facilities. • Store hazardous waste (oils, asbestos fragments, lubricants) safely in sealed, labelled containers. • Train contractors on safe handling of hazardous and construction waste. • Reuse or recycle materials (e.g., pipes, packaging) where feasible. • Keep records of waste volumes and disposal routes. • Ensure appropriate final disposal of all waste streams through certified facilities or authorized export when required.

Potential Environmental Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Production Well Integrity and Water Quality Risks from Monitoring Installation	• DLIs 1 and 5. • Metering • Installation of well monitoring equipment.	If not properly designed/installed, instrumentation can compromise sanitary seals, introduce ingress pathways, create cable abrasion points, obstruct flow, or disturb the well column, leading to transient turbidity, biofouling risk, or interference with pumps/liners.	• Prefer the use of dedicated observation wells to install monitoring devices and avoid interference with production wells. • Maintain or upgrade sanitary seals and use waterproof cable connectors when instruments are installed in production wells. • Disinfect wells before and after installation and ensure that all materials and equipment are suitable for potable water use. • Secure and route cables properly to prevent damage, tangling, or interference with the pump and maintenance tools. • Conduct performance and water quality checks before returning the well to service to confirm safe operation. • Keep updated records of the installation, inspect seals and cables regularly, and plan for maintenance or safe replacement when needed. • Schedule works during periods of low demand and ensure standby supply arrangements to minimize service interruptions.